

Zurich ServiceNow AI Platform Capabilities

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Service Graph Connector for AWS

Use the Service Graph Connector for AWS to securely bring in Amazon Web Services (AWS) data into your ServiceNow instance.

Request apps on the Store

Visit the [ServiceNow Store](#) website to view all the available apps and for information about submitting requests to the store. For cumulative release notes information for all released apps, see the [ServiceNow Store version history release notes](#).

The integration uses AWS native technologies and AWS security best practices to enable cloud teams to connect the data within their ServiceNow workflow. For more information about the Service Graph Connector for AWS, see the [Service Graph Connector for AWS - Introduction](#) article on the ServiceNow Community site.

Supported ServiceNow versions

- Xanadu
- Yokohama
- Zurich

Use cases

The following are examples on how you can use the Service Graph Connector for different ServiceNow applications:

- Visibility into cloud resources, relationships, and state in real time.
- Deep discovery of applications for ITAM or SAM outcomes.
- Governance and compliance outcome.

Important information for upgrading Service Graph Connector for AWS

Before you begin the upgrade process, complete the following tasks:

- If there are any customizations, delete the records associated with the Service Graph Connector for AWS from the Customer Updates [sys_update_xml] table. See [Customer Updates table](#).
- Upgrade any dependencies.

After you've installed the latest version of the Service Graph Connector for AWS:

- Perform a full import of data from your AWS instance. Verify that the **Use last run datetime** field value is cleared for all the AWS data sources in the Data Source [sys_data_source] table.
- Download and rerun the AWS scripts. See [Configure the AWS environment](#).

Configuring a connection for the connector

You can configure a connection for the connector by using the SGC Central view in the CMDB Workspace. The view enables you to discover and install connectors, and then effectively manage the full life cycle of creating, editing, monitoring, and debugging connections. To configure the connector using SGC Central, see [Configure Service Graph Connector for AWS using SGC Central](#).

Important: Unless there are configuration issues, use the SGC Central view in the CMDB Workspace to configure the connection for the connector, as the guided setup method is being deprecated.

CMDB integrations dashboard

The Integration Commons for CMDB store app provides a dashboard with a central view of the status, processing results, and processing errors of all installed integrations. You can see metrics for all integration runs. You can filter the view to a specific CMDB integration, a specific time duration, or a specific integration run. For more details about monitoring AWS integrations in the CMDB Integrations Dashboard, see [Using the CMDB Integrations Dashboard](#).

Data mapping

Data from the AWS data sources is mapped and transformed into the ServiceNow CMDB Configuration Item (CI) class definitions using the

Robust Transform Engine (RTE). Data is inserted into the ServiceNow CMDB using the Identification and Reconciliation Engine (IRE).

Note: If the **Use last run datetime** field for an AWS data source in the Data Source [sys_data_source] table is empty, the connector imports all available initial data. If the **Use last run datetime** field includes a date stamp, the connector imports incremental data that has been newly added since the previous run.

The following table lists the import schedule order, the data sources and import schedules of the same name, the staging tables, the target tables as CMDB CI classes, the import schedule requirement type, and the import schedule dependencies for AWS.

Data mapping for AWS

Order	Name (data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependency
1	SG-AWS-Organization	SG-AWS-Organization [sn_aws_integ_sg_aws_organization]	Cloud Organizations	Required	None
2	SG-AWS-Org-Units	SG-AWS-Org-Units [sn_aws_integ_sg_aws_org_units]	AWS Organizational Unit	Optional	SG-AWS-Organization
3	SG-AWS-Service-Account	SG-AWS-Service-Account [sn_aws_integ_sg_a	Cloud Service Account	Required	SG-AWS-Organization

Order	Name (data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependencies
		ws_service_account]	Cloud Organizations Key Value		
4	SG-AWS-Service-Account-Tags	SG-AWS-Service-Account-Tags [sn_aws_integ_sg_aws_service_account_tags]	Cloud Service Account Key Value	Optional	SG-AWS-Organization SG-AWS-Service-Account
5	SG-AWS-Org-Unit-Accounts	SG-AWS-Org-Unit-Accounts [sn_aws_integ_sg_aws_org_unit_accounts]	Cloud Service Account	Optional	SG-AWS-Organization SG-AWS-Service-Account
6	SG-AWS-Datacenters	SG-AWS-Datacenters [sn_aws_integ_sg_aws_datacenters]	Cloud Service Account AWS Datacenter	Required	SG-AWS-Organization SG-AWS-Service-Account

Order	Name (data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependencies
7	SG-AWS-VPC	SG-AWS-VPC [sn_aws_integ_sg_aws_vpc]	Cloud Service Account Cloud Network AWS Datacenter Key Value SG-AWS Extension Attributes	Required	SG-AWS-Organization SG-AWS-Service-Account SG-AWS-Datacenters
8	SG-AWS-Subnets	SG-AWS-Subnets [sn_aws_integ_sg_aws_subnets]	Availability Zone Cloud Network Cloud Subnet AWS Datacenter Key Value	Required	SG-AWS-Organization SG-AWS-Service-Account SG-AWS-Datacenters SG-AWS-VPC

Order	Name (data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependencies
			SG-AWS Extension Attributes		
9	SG-AWS-Network-Interface	SG-AWS-Network-Interface [sn_aws_integ_sg_aws_network_interface]	Cloud Network Cloud Subnet Cloud Mgmt Network Interface AWS Datacenter Key Value SG-AWS Extension Attributes	Required for a virtual machine (VM) instance	SG-AWS-Organization SG-AWS-Service-Account SG-AWS-Datacenters SG-AWS-VPC
10	SG-AWS-Security-Group	SG-AWS-Security-Group [sn_aws_integ_sg_aws_security_group]	Cloud Network Compute Security Group	Required for a VM instance	SG-AWS-Organization SG-AWS-Service-Account

Order	Name (data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependencies
			AWS Datacenter Key Value SG-AWS Extension Attributes		SG-AWS-Datacenters SG-AWS-VPC
11	SG-AWS-Storage-Volume	SG-AWS-Storage-Volume [sn_aws_integ_sg_aws_storage_volume]	Storage Volume Storage Volume Snapshot AWS Datacenter Key Value SG-AWS Extension Attributes	Required for a VM instance	SG-AWS-Organization SG-AWS-Service-Account SG-AWS-Datacenters SG-AWS-VPC
12	SG-AWS-Image-Private	SG-AWS-Image [sn_aws_integ_sg_aws_image]	Image	Required for a VM instance	SG-AWS-Organization

Order	Name (data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependencies
					SG-AWS-Service-Account SG-AWS-Datacenters SG-AWS-VPC
13	SG-AWS-Image-Id	SG-AWS-Image-Id [sn_aws_integ_sg_aws_image_id]	Image	Required for a VM instance	SG-AWS-Organization SG-AWS-Service-Account SG-AWS-Datacenters SG-AWS-VPC
14	SG-AWS-Hardware-Type	SG-AWS-Hardware-Type [sn_aws_integ_sg_aws_hardware_type]	Hardware Type	Required for a VM instance	SG-AWS-Organization SG-AWS-Service-Account

Order	Name (data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependencies
			AWS Datacenter		SG-AWS-Datacenters SG-AWS-VPC
15	SG-AWS-EC2	SG-AWS-EC2 [sn_aws_integ_sg_aws_ec2]	Virtual Machine Instance The following CIs are populated when populating the Virtual Machine Instance CI: Server VNIC Endpoint Storage Mapping Block Endpoint	Optional	SG-AWS-Organization SG-AWS-Service-Account SG-AWS-Datacenters SG-AWS-VPC SG-AWS-Subnets SG-AWS-Network-Interface SG-AWS-Hardware-Type

Order	Name (data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependencies
			IP Address Network Adapter Key Value		SG-AWS-Security-Group SG-AWS-Storage-Volume SG-AWS-Image-Private SG-AWS-Image-Id
16	SG-AWS-ELB-V1	SG-AWS-ELB-V1 [sn_aws_integ_sg_aws_elb_v1]	Cloud Load Balancer Compute Security Group Availability Zone AWS Datacenter Key Value	Optional	SG-AWS-Organization SG-AWS-Service-Account SG-AWS-Datacenters

Order	Name (data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependencies
			SG-AWS Extension Attributes		
17	SG-AWS-ELB-V2	SG-AWS-ELB-V2 [sn_aws_integ_sg_aws_elb_v2]	Cloud Load Balancer Compute Security Group Availability Zone AWS Datacenter Key Value SG-AWS Extension Attributes	Optional	SG-AWS-Organization SG-AWS-Service-Account SG-AWS-Datacenters
18	SG-AWS-RDS	SG-AWS-RDS [sn_aws_integ_sg_aws_rds]	Cloud DataBase AWS Datacenter	Optional	SG-AWS-Organization SG-AWS-Service-Account

Order	Name (data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependencies
			Key Value SG-AWS Extension Attributes		SG-AWS-Datacenters
19	SG-AWS-API-Gateway	SG-AWS-API-Gateway [sn_aws_integ_sg_aws_api_gateway]	Cloud Gateway [cmdb_ci_cloud_gateway] AWS Datacenter Key Value SG-AWS Extension Attributes	Optional	SG-AWS-Organization SG-AWS-Service-Account SG-AWS-Datacenters
20	SG-AWS-Lambda	SG-AWS-Lambda [sn_aws_integ_sg_aws_lambda]	Cloud Function	Optional	SG-AWS-Organization SG-AWS-Service-Account

Order	Name (data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependencies
					SG-AWS-Datacenters
21	SG-AWS-S3	SG-AWS-S3 [sn_aws_integ_sg_aws_s3]	Cloud Object Storage	Optional	SG-AWS-Organization SG-AWS-Service-Account SG-AWS-Datacenters
22	SG-AWS-DynamoDB	SG-AWS-DynamoDB [sn_aws_integ_sg_aws_dynamodb]	DynamoDB Table AWS Datacenter Key Value SG-AWS Extension Attributes	Optional	SG-AWS-Organization SG-AWS-Service-Account SG-AWS-Datacenters

Order	Name (data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependencies
23	SG-AWS-Software-Inventory	SG-AWS-Software-Inventory [sn_aws_integ_sg_aws_software_inventory] SG-AWS-Software-Staging [sn_aws_integ_sg_aws_temp_software_staging]	When the Software Asset Management (SAM) application isn't installed: Software Packages Software Instance Server When the SAM application is installed: Software Installation Server	Optional	SG-AWS-Organization SG-AWS-Service-Account SG-AWS-Datacenters SG-AWS-VPC SG-AWS-Subnets SG-AWS-Network-Interface SG-AWS-Hardware-Type SG-AWS-Security-Group SG-AWS-Storage-Volume

Order	Name (data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependencies
					SG-AWS-Image-Private SG-AWS-Image-Id SG-AWS-EC2
24	SG-AWS-Software-Remove	SG-AWS-Software-Remove [sn_aws_integ_sg_aws_software_remove]	None	Optional	SG-AWS-Organization SG-AWS-Service-Account SG-AWS-Datacenters SG-AWS-VPC SG-AWS-Subnets SG-AWS-Network-Interface

Name					
Order	(data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependency
					SG-AWS-Hardware-Type SG-AWS-Security-Group SG-AWS-Storage-Volume SG-AWS-Image-Private SG-AWS-Image-Id SG-AWS-EC2 SG-AWS-Software-Inventory
25	SG-AWS-SSM-SendCommand	SG-AWS-SSM-SendCommand [sn_aws_integ_sg_aws_ssm_sendcommand]	Application Running Process [cmdb_running_process]	Optional	SG-AWS-Organization SG-AWS-Service-Account

Order	Name (data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependencies
			TCP Connections [cmdb_tcp]		SG-AWS-Datacenters SG-AWS-VPC SG-AWS-Subnets SG-AWS-Network-Interface SG-AWS-Security-Group SG-AWS-Storage-Volume SG-AWS-Image-Private SG-AWS-Image-Id SG-AWS-EC2
26	SG-AWS-Tags	SG-AWS-Tags [sn_aws_in]	DynamoDB Table	Optional	SG-AWS-Organization

Order	Name (data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependencies
		teg_sg_aws_tags]	Cloud Load Balancer Cloud Function Key Value		SG-AWS-Service-Account SG-AWS-Datacenters SG-AWS-ELB-V1 SG-AWS-ELB-V2 SG-AWS-Dynamodb SG-AWS-Lambda
27	SG-AWS-VM-Hw-Consolidation	SG-AWS-VM-Hw-Consolidation [sn_aws_in teg_sg_aws_vm_hw_consolidation]	Virtual Machine Instance Server	Optional	SG-AWS-Organization SG-AWS-Service-Account SG-AWS-Datacenters

Order	Name (data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependencies
					SG-AWS-VPC SG-AWS-Subnets SG-AWS-Network-Interface SG-AWS-Security-Group SG-AWS-Storage-Volume SG-AWS-Image-Private SG-AWS-Image-Id SG-AWS-EC2 SG-AWS-Hardware-Type
28	SG-AWS-EKS-Cluster	SG-AWS-EKS-Cluster [sn_aws_in	Kubernetes Cluster	Optional	SG-AWS-Organization

Name					
Order	(data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependencies
		teg_sg_aws_eks_cluster]	AWS Datacenter Key Value SG-AWS Extension Attributes		SG-AWS-Service-Account SG-AWS-Datacenters SG-AWS-VPC SG-AWS-Subnets SG-AWS-Network-Interface SG-AWS-Security-Group SG-AWS-Storage-Volume SG-AWS-Image-Private SG-AWS-Image-Id SG-AWS-EC2

Order	Name (data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependencies
					SG-AWS-Hardware-Type SG-AWS-VM-Hw-Consolidation
29	SG-AWS-EKS-Cluster-2	SG-AWS-EKS-Cluster-2 [sn_aws_integ_sg_aws_eks_cluster_2]	Kubernetes Cluster AWS Datacenter Key Value SG-AWS Extension Attributes	Optional	SG-AWS-Organization SG-AWS-Service-Account SG-AWS-Datacenters SG-AWS-VPC SG-AWS-Subnets SG-AWS-Network-Interface SG-AWS-Security-Group

Order	Name (data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependencies
					SG-AWS-Storage-Volume SG-AWS-Image-Private SG-AWS-Image-Id SG-AWS-EC2 SG-AWS-Hardware-Type SG-AWS-VM-Hw-Consolidation SG-AWS-EKS-Cluster
30	SG-AWS-EKS-FULL	SG-AWS-EKS-FULL [sn_aws_integ_sg_aws_eks_full]	Kubernetes Cluster Server Kubernetes	Optional	SG-AWS-Organization SG-AWS-Service-Account

Name					
Order	(data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependency
			Namespa ce		SG-AWS-Datacent ers
			Kubernete s Node		SG-AWS-VPC
			Kubernete s Service		SG-AWS-Subnets
			Kubernete s Pod		SG-AWS-Network-Interface
			Docker Container		SG-AWS-Security-Group
			Docker Image		SG-AWS-Storage-Volume
			Kubernete s Volume		SG-AWS-Image-Private
			Kubernete s Deployme nt		SG-AWS-Image-Id
			Kubernete s DaemonS et		SG-AWS-EC2
			Kubernete s ReplicaSet		SG-AWS-Hardware-Type

Order	Name (data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependencies
					SG-AWS-VM-Hw-Consolidation SG-AWS-EKS-Cluster SG-AWS-EKS-Cluster-2
31	SG-AWS-Generic-Resources	SG-AWS-Generic-Resources [sn_aws_integ_sg_aws_generic_resources]	Cloud Resource SG-AWS Extension Attributes	Optional	SG-AWS-Organization
32	SG-AWS-Redshift-Cluster	SG-AWS-Redshift-Cluster [sn_aws_integ_sg_aws_redshift_cluster]	Amazon Redshift	Optional	SG-AWS-Organization SG-AWS-Service-Account SG-AWS-Datacenters

Order	Name (data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependencies
33	SG-AWS-Get-Inventory	SG-AWS-Get-Inventory [sn_aws_in teg_sg_a ws_get_in ventory]	Server	Required	SG-AWS-Organization SG-AWS-Service-Account SG-AWS-Datacenters
34	SG-AWS-GenericTags	SG-AWS-GenericTags [sn_aws_in teg_sg_a ws_generi ctags]	Cloud Resource Key Value	Optional	SG-AWS-Organization SG-AWS-Generic-Resources
35	SG-AWS-SendCommand	SG-AWS-SendCommand [sn_aws_in teg_sg_a ws_ssm_se ndcomm and]	None	Optional	SG-AWS-Organization SG-AWS-Service-Account SG-AWS-Datacenters

Name					
Order	(data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependencies
			Note: The SG-AWS-Send Command data source doesn't have target CMDB CI classes. This data source populates the data into the sn_aws_integ_sg_aws_sendcommand staging table, but the import records		SG-AWS-VPC SG-AWS-Subnets SG-AWS-Network-Interface SG-AWS-Security-Group SG-AWS-Storage-Volume SG-AWS-Image-Private SG-AWS-Image-Id SG-AWS-EC2

Order	Name (data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependencies
36	SG-AWS-SSM-GetS3Object	SG-AWS-SSM-GetS3Object [sn_aws_integ_sg_aws_ssm_gets3object]	Server Running Process [cmdb_running_process] TCP Connections [cmdb_tcp]	Optional	SG-AWS-Organization SG-AWS-Service-Account SG-AWS-Datcenters SG-AWS-VPC SG-AWS-Subnets SG-AWS-Network-Interface SG-AWS-Security-Group SG-AWS-Storage-Volume SG-AWS-Image-Private

Order	Name				
	(data source or import schedule)	Staging table	CMDB CI classes	Import schedule requirement type	Import schedule dependency
					SG-AWS-Image-Id
					SG-AWS-EC2
					SG-AWS-SendCommand

- The SG-AWS-Generic-Resources data source imports data for generic resources that aren't tracked by other data sources. The connector uses the Service Graph Resource Inclusion Whitelist [sn_cmdb_int_util_service_graph_resource_inclusion_whitelist] table to differentiate between generic and other supported resource types.

The connector first populates all supported resources in the Service Graph Resource Inclusion Whitelist [sn_cmdb_int_util_service_graph_resource_inclusion_whitelist] table. These resources, categorized under their respective supported resource types, have specific data sources designated for ingestion within the connector. When the SG-AWS-Generic-Resources data source is executed and retrieves unsupported resource types, they are added to the Service Graph Resource Inclusion Whitelist [sn_cmdb_int_util_service_graph_resource_inclusion_whitelist] table and categorized as generic.

- To import global generic resources such as IAM user and IAM group, specify a standard AWS region that has Config enabled with includeGlobalResourceTypes set to true by updating the value of the sn_aws_integ.global_generic_resource_region system property for the SG-AWS-Generic-Resources data source.

If an aggregator is configured, and the sn_aws_integ.global_generic_resource_region value is not specified,

the aggregator region is assigned as the value of this system property for importing global generic resources.

- If the AWS Systems Manager (SSM) service isn't enabled, the connector populates the server records in the Server [cmdb_ci_server] class. If the AWS SSM service is enabled, then based on the platform type obtained through the SSM service, the server records are populated in either the Linux Server [cmdb_ci_linux_server] class or the Windows Server [cmdb_ci_win_server] class. The Server [cmdb_ci_server] class is the parent class of the Linux Server [cmdb_ci_linux_server] and the Windows Server [cmdb_ci_win_server] classes.
- All labels associated with an AWS resource are added to the Key Value [cmdb_key_value] table.

Note: You can use the [CMDB Data Manager](#) to delete tag data from retired CIs in the Key Value [cmdb_key_value] table based on conditions like retention time and discovery source. A scheduled job runs the policy, which can be configured to execute during off-peak hours.
- The basic information about an AWS resource is stored in the SG-AWS Extension Attributes [sn_aws_extension_attributes] table.
- Starting with the Service Graph Connector for AWS 2.10.0 version, the SG-AWS-Get-Inventory data source runs before the SG-AWS-EC2 data source and creates a Server [cmdb_ci_server] CI with the host name mapped to the Name attribute, instead of being mapped to the VM name.
- The SG-AWS-GenericTags data source imports tag data only for generic resources that have an ARN key. You can use the SG-AWS Extension Attributes [sn_aws_extension_attributes] table to verify which generic resources have an ARN key.

For more information on where data is saved when pulling data from AWS, see [CMDB classes targeted in Service Graph Connector for AWS](#) and [Supported AWS resource types](#).

When you run the diagnostic test, the data is loaded in the following tables:

- SG AWS Diagnostic Details [sn_aws_integ_sg_aws_diagnostic_details]

- SG-AWS Diagnostic Summary
[sn_aws_integ_sg_aws_diagnostic_summary]
- SG AWS Diagnostic Summary Notes
[sn_aws_integ_sg_aws_diagnostic_summary_notes]

You can use the IntegrationHub ETL app to view the data maps. See [IntegrationHub ETL](#) for more information.

The AWS configuration data for each connection is stored in the SG AWS Application properties [sn_aws_integ_sg_aws_application_properties] table.

For more information about how CI information is pulled from AWS, see the [Service Graph Connector for AWS - Functional Spec and CI](#) article on the ServiceNow Community site.

BYOL support

Bringing your own licenses (BYOL) is the process of bringing previously purchased on-premises licenses to Amazon AWS Cloud. The connector supports the BYOL of the Image CIs for Oracle Database servers on Amazon AWS Cloud by populating the Key Value [cmdb_key_value] table with the licensing information. When the licensing information is found, the key is set to windows_OS_License_Type_automatic and the key value is set to BYOL in the Key Value [cmdb_key_value] table.

To support BYOL, the sn_aws_integ.load_all_images system property is enabled during the initial pull to import the metadata of all Image CIs included in the SG-AWS-Image-Id data source. After the initial pull, the property is deactivated automatically. The SG-AWS-Image-Id data source then imports the metadata of the Image CIs without names only.

Event-based discovery

The Service Graph Connector for AWS works with event-based discovery. Events that are pulled from AWS can create and update CIs that were brought by the Service Graph Connector for AWS.

Managing retired CIs in EKS components

If the Amazon Elastic Kubernetes Services (EKS) CIs are deleted in EKS components such as pods, services, and volumes, the corresponding CI entries are automatically set to **Retired** in the following CMDB CI classes:

Kubernetes Cluster, Kubernetes Node, Kubernetes Pod, Kubernetes Service, Kubernetes DaemonSet, Kubernetes Namespace, Kubernetes Deployment, Docker Container, Kubernetes Volume

Shared VPC and subnets support

Import CI relationships established between virtual private clouds (VPCs), subnets, and network interfaces hosted in different AWS accounts. For example, a VPC in one AWS account can be connected to a subnet in the same account, which can then link to a network interface and a virtual machine in another AWS account.

Additional resources

See the following articles on the ServiceNow Community site for any additional information on the AWS set up:

- [Service Graph Connector for AWS - Introduction](#)
- [How do I configure the AWS Service Graph Connector?](#)
- [Service Graph Connector for AWS - Diagnostic Tool & Troubleshooting Issues](#)
- [Service Graph Connector for AWS - Functional Spec and CI](#)
- [Cloud Discovery and SG-AWS](#)

Related content

[Service Graph Connectors](#)

[Enable optimized deletion of retired CIs in AWS](#)

[Enable optimized SG-AWS-SendCommand data source performance](#)

[Handling import failures for large AWS resources](#)

[Handling memory utilization errors for Service Graph Connector for AWS](#)

[Life cycle management of records in Service Graph Connector for AWS](#)

[Record removal process in Service Graph Connector for AWS](#)

Configure the AWS environment

Configure your Amazon Web Services (AWS) environment to import data using the Service Graph Connector for AWS.

Before you begin

Install the Service Graph Connector for AWS application (sn_aws_integ). For more information, see [Install a ServiceNow Store application](#).

Role required: application administrator

Procedure

1. [Learn about the AWS resources used by the connector](#).
Get familiar with the AWS concepts to learn how the Service Graph Connector for AWS is integrated with AWS.
2. [Download the AWS scripts](#).
Download the AWS scripts included within the guided setup of the Service Graph Connector for AWS.
3. [Review the requirements before executing the AWS scripts](#).
Complete the prerequisites before you execute the AWS scripts.
4. [Execute the AWS scripts](#).
Execute scripts provided with the Service Graph Connector for AWS to set up the AWS environment for importing data.

AWS resources used by the Service Graph Connector for AWS

Get familiar with the AWS concepts to learn how the Service Graph Connector for AWS is integrated with Amazon Web Services (AWS).

AWS Config service and configuration recorder

Important: The AWS Config service and AWS configuration recorder are required for setting up the connector.

The AWS Config service monitors and records changes to your AWS resource configurations.

The AWS configuration recorder detects changes in resource configurations and captures these changes as configuration items (CIs). This is required for setting up the connector. The configuration recorder enables recording all hardware data in AWS Config. See [What Is AWS Config?](#) on the AWS Documentation site.

The Service Graph Connector for AWS includes the `EnableAWSConfig.yml` script to enable the AWS Config service that instead enables the configuration recorder. See [Executing scripts required for setting up AWS](#).

Note: Ensure that the AWS Config service is enabled for all applicable AWS accounts and regions.

AWS Config aggregator

Important: The AWS Config aggregator is optional for setting up the connector.

The AWS Config aggregator collects the AWS Config configuration and compliance data from the following sources:

- Multiple accounts and multiple regions
- Single account and multiple regions
- An organization in AWS organizations and all the accounts within the organization that have AWS Config enabled.

The advantages of using an AWS Config aggregator with the Service Graph Connector for AWS are:

- Gets all the data from a single location.
- Gets the bootstrap updates (baseline configurations) and the incremental updates (new configurations added after the last update).
- Doesn't require looping into each account and region.
- Accelerates pulling data.

Due to these advantages, consider leveraging the AWS Config aggregator for pulling data from multiple accounts or multiple regions.

Note: For detecting any deleted resources, the connector uses the `config:ListDiscoveredResources` API to loop through each AWS account and region and update the CMDB CI accordingly. As a date range for selecting resources can't be specified in the `ListDiscoveredResources` API, the connector might make multiple API calls to gather all the data that might impact the performance of the connector.

For more information on setting up an AWS Config aggregator, see [Multi-Account Multi-Region Data Aggregation](#) and [Setting Up an Aggregator Using the Console](#) on the AWS Documentation site.

AWS Systems Manager and AWS Systems Manager Inventory

Important: AWS Systems Manager and AWS Systems Manager Inventory are required for setting up the deep discovery feature.

The AWS Systems Manager enables fetching server data, also called as deep discovery data, from EC2 instances across AWS accounts and regions through SSM documents. The deep discovery data includes host name, serial number, CPU data, TCP data, and process information.

The AWS Systems Manager Inventory imports the software data installed on the EC2 instances. The Inventory resource group in AWS Systems Manager collects information about the EC2 instances and the software applications installed on them.

Ensure that the following items are configured in all AWS accounts:

- The AWS Systems Manager Agent (SSM Agent) is installed on all managed EC2 instances.
- The `AmazonSSMForInstancesRole` IAM instance profile role is attached as the instance profile on EC2 instances.
- The AWS Systems Manager Inventory is configured in each AWS region.
- The AWS Systems Manager has access to the managed EC2 instances.

Note: By default, AWS Systems Manager doesn't have permission to perform actions on EC2 instances. You can grant access by attaching the AmazonSSMForInstancesRole IAM instance profile role to the EC2 instance. See [Setting up AWS Systems Manager](#) on the AWS Documentation site.

The advantages of using AWS Systems Manager and AWS Systems Manager Inventory are:

- The AWS Systems Manager enables getting the detailed server data such as host name, serial number, CPU data, TCP data, and process information.
- The AWS Systems Manager Inventory enables the server classification and getting the software data.

Download the AWS scripts

Download the AWS scripts included within the guided setup of the Service Graph Connector for AWS.

Before you begin

- Ensure that you have installed the Service Graph Connector for AWS application (sn_aws_integ). For more information, see [Install a ServiceNow Store application](#).
- Ensure that you've selected the **Service Graph Connector for AWS** application scope by using the application picker. See [Application picker](#).

Role required: admin

Procedure

1. Navigate to **All > Service Graph Connector for AWS > Setup**.
2. On the Getting started page, select **Get Started**.
3. In the Configure the AWS environment section of the Service Graph Connector for AWS page, select **Get Started**.
4. For the Download the scripts task, select **Configure**.

5. When you download all the scripts onto your local machine, select **Close** to close the Download the scripts dialog box and return to the guided setup.
6. Set the Download the scripts task to complete by selecting **Mark as Complete**.

Prerequisites for executing scripts

Complete the prerequisites before you execute the AWS scripts.

Important: Ensure that you have downloaded the scripts available within the Service Graph Connector for AWS. See [Download the AWS scripts](#).

Decide the following details to be used later during the execution of the AWS scripts:

- [ServiceNow IAM role](#)
- [ServiceNow IAM username](#)
- [S3 bucket for the ServiceNow IAM role](#)

Note: Define an S3 bucket only when you are using [deep discovery scripts](#).

Determine the ServiceNow IAM role

Determine the identity and access management (IAM) role that performs read-only operations in member accounts to fetch the configuration items (CIs) from the AWS environment.

By default, the `CreateSnowOrganizationAccountAccessRoleInMemberAccount.yml` script creates the `SnowOrganizationAccountAccessRole` IAM role. You can use the default name created by the script or create a new IAM role. However, when required as an input parameter, you must enter the same IAM role across scripts. See [Executing scripts required for setting up AWS](#).

Determine the ServiceNow IAM username

Determine the name of the IAM user that assumes the ServiceNow IAM role in the member accounts.

By default, the `CreateServiceNowUser.yml` script creates the `NOWSGCUser` IAM user. You can use the default name created by the script or create a new IAM user. However, when required as an input parameter, you must enter the same IAM username across scripts. See [Executing scripts required for setting up AWS](#).

Define the S3 bucket for deep discovery

Set up an S3 bucket with read and delete privileges for the ServiceNow IAM role to store and delete the `SendCommand` API responses when importing AWS data.

Before you begin

Role required: Application administrator

About this task

Create an S3 bucket for the Service Graph Connector for AWS application and enable the ServiceNow IAM role to have access to this bucket in the organization.

Note: Use an S3 bucket only when you want to perform deep discovery on EC2 instances.

Procedure

1. Create an S3 bucket in an AWS account region.
See [Creating a bucket](#) on the AWS documentation site.

Note: The S3 bucket must have the following permission settings.

S3 bucket permissions and their settings

Permission	Setting
Access	Bucket and objects not public
S3 Block Public Access	Block public access to S3 buckets and objects

For more information, see [S3 Block Public Access](#) on the AWS documentation site.

2. Add a bucket policy.

See [Bucket policies](#) on the AWS documentation site.

To access the S3 bucket you created in [step 1](#), its bucket policy must allow the IAM instance profile role attached to the managed EC2 instances. You can either create a bucket policy or grant access to your AWS member account in the bucket access control list (ACL). The member account must include the EC2 instances.

Note: Adding an AWS member account to the bucket ACL allows all users and roles in the member account to access the S3 bucket.

Refer to the following sample code when adding a bucket policy.

```
{
  "Version": "2012-10-17",
  "Id": "S3PolicyforServiceNowIAMrole",
  "Statement": [
    {
      "Sid": "EC2S3Access",
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::AWS-ACCOUNT:role/INSTANCE-PROFILE-ROLE-NAME"
      },
      "Action": [
        "s3:GetObject",

```



```

        "s3:PutObject",
        "s3:PutObjectAcl"
    ],
    "Resource": "arn:aws:s3::DOC-EXAMPLE-BUCKET/*"
}
}

```

Where

- SOURCE-AWS-ACCOUNT is the AWS account ID of the member account that includes the EC2 instances.
- INSTANCE-PROFILE-ROLE-NAME is the IAM instance profile that is attached to the EC2 instances.

By default, the `AmazonSSMForInstancesRoleSetup.yml` script creates the `AmazonSSMForInstancesRole` IAM instance profile role and attaches the role to the `AmazonSSMManagedInstanceCore` bucket policy. See [Executing scripts required for setting up AWS](#).

- DOC-EXAMPLE-BUCKET is the S3 bucket name.

The following example bucket policy shows the effect, principal, action, and resource elements. The policy allows `AmazonSSMRoleForInstances`, an IAM instance profile role in an account with ID `123456789000`, `s3:GetObject`, `s3:PutObject`, and `s3:PutObjectAcl` S3 permissions on the `myS3Bucket` bucket.

```

{
  "Version": "2012-10-17",
  "Id": "S3PolicyforServiceNowIAMrole",
  "Statement": [
    {
      "Sid": "EC2S3Access",
      "Effect": "Allow",
      "Principal": {
        "AWS": [
          "arn:aws:iam::123456789000:role/AmazonSSMRoleForInstances",
          "arn:aws:iam::123456789001:role/AmazonSSMRoleForInstances",
          "arn:aws:iam::123456789002:role/AmazonSSMRoleForInstances",

```

```

        "arn:aws:iam::123456789003:role/AmazonSSMRoleForInstances",
        "arn:aws:iam::123456789004:role/AmazonSSMRoleForInstances"
    ],
    "Action": [
        "s3:GetObject",
        "s3:PutObject",
        "s3:PutObjectAcl"
    ],
    "Resource": "arn:aws:s3:::myS3Bucket/*"
}

```

3. Attach IAM permissions to the instance profile role for EC2 instances to publish the SendCommand API responses to the S3 bucket you created in [step 1](#).
See [Using instance profiles](#) and [Attach an IAM role to an instance](#) on the AWS documentation site.
The IAM instance profile role attached to the managed EC2 instances must have the `s3:GetObject`, `s3:PutObject`, and `s3:PutObjectAcl` S3 permissions to allow access to the S3 bucket as shown in the following sample policy.

```

{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublishTerminalOutputToS3",
      "Effect": "Allow",
      "Action": [
        "s3:PutObject",
        "s3:GetObject",
        "s3:PutObjectAcl"
      ],
      "Resource": "arn:aws:s3:::DOC-EXAMPLE-BUCKET/*"
    }
  ]
}

```

Where, DOC-EXAMPLE-BUCKET is the S3 bucket name.

Note: Make sure you add the suffix `/*` at the end of the bucket name to enable creating files under the bucket name.

Executing scripts required for setting up AWS

You must execute scripts provided with the Service Graph Connector for AWS to set up the AWS environment for importing data.

Important: Before executing an AWS script, ensure that you have completed the prerequisites. See [Prerequisites for executing scripts](#).

The AWS scripts provided with the connector configure AWS resources to import the configuration items (CIs) data into the CMDB. To learn more, see [AWS resources used by the Service Graph Connector for AWS](#).

Based on the AWS environment requirements, the scripts provided with the Service Graph Connector for AWS are categorized as described in the following table.

Basic scripts



Scripts for configuring the AWS environment to import data by using the connector.

Deep discovery scripts



Scripts for setting up deep discovery on Amazon Elastic Compute Cloud (Amazon EC2) instances.

Amazon EKS scripts



Scripts for setting up Amazon Elastic Kubernetes Service (EKS) clusters.

Basic scripts

Use the basic scripts to configure the AWS environment for importing data using the Service Graph Connector for AWS.

The following table describes the basic AWS scripts available with the connector, the input parameters entered when executing a script, the conditions to execute the scripts, and the script execution results.

Basic script details

Script	Input parameters	Execution condition	Result
EnableAWSConfig.yml	None	Execute the script in all the AWS accounts and AWS regions by creating a CloudFormation StackSet in the management account. See Create a stack set on the AWS documentation site.	Enables the AWS Config recorder.
CreateServiceNowUser.yml	SNUserName Name of the ServiceNow IAM user that was created as part of the setup. See Prerequisites for executing scripts. Default value: NOWSGCUser	Execute the script by creating a stack either in the management account or in a designated member account. See Creating a stack on the AWS CloudFormation	Creates the ServiceNow IAM user.

Script	Input parameters	Execution condition	Result
	MbrActRoleName Name of the ServiceNow IAM role that was created as part of the setup. See Prerequisites for executing scripts. Default value: SnowOrganizationAccountAccessRole	console on the AWS documentation site.	
CreateSnowOrganizationAccountAccessRoleInMemberAccount.yml	ACNNBR Management account ID when the ServiceNow IAM user is in a management account Or Designated member account ID when the ServiceNow IAM user is in a designated member account.	Execute the script in all the AWS accounts by creating a CloudFormation StackSet in the management account. See Create a stack set on the AWS documentation site.	Enables read-only IAM policies, roles, and groups for the ServiceNow IAM user.

Script	Input parameters	Execution condition	Result
	S3Bucket Amazon S3 bucket name to get the SendCommand output. ServiceNowUserName Name of the ServiceNow IAM user that was created as part of the setup. See Prerequisites for executing scripts. Default value: <code>NOwSGCUser</code>		
SnowDesignatedAccountAccessRoleInManagementAccount.yml	MEMBERACTNBR Member account ID where the ServiceNow IAM user was created.	Execute the script by creating a stack in the management account. See Creating a stack on the AWS CloudFormation console on the AWS documentation site.	Creates the ServiceNow IAM role in the management account.

Script	Input parameters	Execution condition	Result
		Note: Use the SnowDesignatedAccountAccessRoleInManagementAccount.yml script only when the ServiceNow IAM user was created in a member account.	
AWS-SystemsManager-AutomationExecutionRole.yml	None	Execute the script in all the AWS accounts by creating a CloudFormation StackSet in the management account. See Create a stack set on the AWS documentation site.	Provisions the execution role necessary to run automations in member accounts. A prerequisite for configuring Systems Manager Automation.
AWS-SystemsManager-AutomationAdminRole.yml	None	Execute the script by creating a stack in the	Provisions the administrator role in the management account

Script	Input parameters	Execution condition	Result
inistrationRole.yml		management account. See Creating a stack on the AWS CloudFormation console on the AWS documentation site.	necessary to run cross-account automation across multiple accounts. A prerequisite for configuring Systems Manager Automation.

Deep discovery scripts

Use the deep discovery scripts to set up deep discovery on Amazon EC2 instances.

Note: Execute the deep discovery scripts only when you want to perform deep discovery on EC2 instances.

The following table describes the deep discovery scripts, the input parameters entered when executing a script, the conditions to execute the scripts, and the script execution results.

Deep discovery script details

Script	Input parameters	Execution condition	Result
AmazonSSMForInstancesRoleSetup.yml	S3Bucket S3 bucket name that collects the details from EC2 instances. See Prerequisites for executing scripts .	Execute the script in all the AWS accounts by creating a CloudFormation StackSet in the management account. See Create a stack set	Creates the AmazonSSMForInstancesRole IAM instance profile role to be attached to the EC2 instances.

Script	Input parameters	Execution condition	Result
		on the AWS documentation site.	
SG-AWS-RunShellScript-Setup.yml	None	Execute the script in all the AWS accounts and the AWS regions by creating a CloudFormation StackSet in the management account. AWS administrators must update SSM documents and verify that EC2 instances can execute relevant commands for proper integration. See Create a stack set on the AWS documentation site.	Creates AWS Systems Manager (SSM) documents to fetch deep discovery data from a Linux EC2 instance. Retrieves version details for middleware applications, including Apache HTTP server, Nginx server, Apache Tomcat server, and MySQL instance.
SG-AWS-RunPowerShellScript-Setup.yml	None	Execute the script in all the AWS accounts and the AWS	Creates AWS SSM documents to fetch deep discovery data

Script	Input parameters	Execution condition	Result
		regions by creating a CloudFormation StackSet in the management account. See Create a stack set on the AWS documentation site.	from a Windows EC2 instance.

Amazon EKS scripts

Use the Amazon EKS scripts to set up Amazon Elastic Kubernetes Service (EKS) clusters.

Note: Execute the Amazon EKS scripts only when the Amazon EKS service for Kubernetes clusters is required.

The following table describes the Amazon EKS scripts, the conditions to execute the scripts, and the script execution results.

Amazon EKS script details

Script	Execution condition	Result
SG-AWS-RunKubeCtlEKSNAMES ShellScript.yml	Execute the script in all the AWS accounts and the AWS regions where the EC2 Bastion hosts are located by creating a CloudFormation StackSet in	Creates an AWS SSM document to discover EKS clusters associated with EC2 Bastion hosts.

Script	Execution condition	Result
	the management account. See Create a stack set on the AWS documentation site.	Note: An AWS Cloud administrator can update the SSM document in their AWS setup.
SG-AWS-RunKubeCtlShellScript.yml	Execute the script in all the AWS accounts and the AWS regions where the EC2 Bastion hosts are located by creating a CloudFormation StackSet in the management account. See Create a stack set on the AWS documentation site.	Creates an AWS SSM document to fetch CIs related to Kubernetes components, such as pods, services, and deployments, from EKS clusters. Note: An AWS Cloud administrator can update the SSM document in their AWS setup.

Configure Service Graph Connector for AWS using SGC Central

Use the playbook available with the SGC Central application to set up the Service Graph Connector for AWS for pulling in AWS data into the CMDB

Before you begin

- Install Service Graph Connector for AWS version 2.7.0 or later from the ServiceNow Store. For ServiceNow Store installation steps, see [Install a ServiceNow Store application](#).
- Verify that you've completed the prerequisites for setting up AWS. See [Configure the AWS environment](#).

Role required: The following table shows the roles required for each stage of the playbook.

Role required for each playbook stage

Stage	Role
Prerequisites	admin
Setup	cmdb_inst_admin or admin

About this task

The playbook experience for onboarding connectors is activated with SGC Central in the CMDB Workspace. To configure the SGC Central application, see [Configuring SGC Central](#) and for more information on how to interact with a playbook, see [Interact with Playbook](#).

Procedure

1. Navigate to **Workspaces > CMDB Workspace**.
2. In the CMDB Workspace, select **SGC Central**.
3. On the Dashboard page, select **Create connection**.

Tip: Alternatively, you can select **Create connection** on the All connections page.
4. On the Create connection window, select the AWS connector type and then select **Create connection**.
5. Complete the initial prerequisites when setting up a connection for the first time using a connector.

Note: This step is required only during the first-time setup. See [Perform initial setup tasks when creating a connection in SGC Central](#).
6. Complete the prerequisites for setting up the AWS environment.
 - a. Configure the basic setup in the AWS environment required for importing data using the connector.

- a. In the **Prerequisites** stage of the playbook, select the **Download basic scripts** activity.
 - b. Execute the scripts to configure the AWS environment.

For more information on executing scripts, see [Basic scripts](#).
 - c. Select the **I have read the instructions and executed the script accordingly** check box to confirm that you have executed the scripts.
 - d. Select **Continue**.
- b. Set up deep discovery on Amazon Elastic Compute Cloud (Amazon EC2) instances.

Note: Complete this step only to perform deep discovery on EC2 instances. Else, select **Skip**.
 - a. In the **Prerequisites** stage of the playbook, select the **Download deep discovery scripts** activity.
 - b. Execute the scripts to configure Amazon EC2 instances for deep discovery.

For more information on executing scripts, see [Deep discovery scripts](#).
 - c. Select **Continue**.
- c. Set up Amazon Elastic Kubernetes Service (EKS) clusters.

Note: Complete this step only when the Amazon EKS service for Kubernetes clusters is required. Else, select **Skip**.
 - a. In the **Prerequisites** stage of the playbook, select the **Download Amazon EKS scripts** activity.
 - b. Execute the scripts to set up Amazon EKS clusters.

For more information on executing scripts, see [Amazon EKS scripts](#).
 - c. Select **Continue**.

7. Complete the setup for configuring the connector for importing data.

a. Create and test connection.

a. In the **Setup** stage of the playbook, select the **Create and test connection** activity.

b. On the form, fill in the fields.

Create and test connection form

Field	
Connection name	Name to identify the AWS connection record. For example, SG_AWS_CredentialAlias_Org.
Access Key ID	Access key ID of the IAM user that has permissions to interact with the AWS resources.
Secret Access Key	Secret access key that corresponds to the Access key ID required for authenticating the connection securely.
Use MID Server	Option to use a MID Server. Note: Use of a MID Server is optional.
Mid Selection	Name of the MID Server used by the connector. This field appears only when the Use MID Server check box is selected.

- c. Select **Create and test connection**.
- d. Once the connection test is complete, select **Continue**.
- b. Set configuration properties for the connection to access the AWS resources.
 - a. In the **Setup** stage of the playbook, select the **Set configuration properties** activity.
 - b. In the Organization details section, fill in the organization details including the account identifier, name and description of the AWS organization.
 - c. In the S3 account details section, fill in the details.

S3 account details

Field	Description
S3 account ID	Numeric identifier of the AWS account that hosts the Amazon Simple Storage Service (Amazon S3) bucket.
S3 bucket name	Name of the Amazon S3 bucket that collects the details from Amazon EC2 instances.
S3 region	Region where the Amazon S3 bucket resides.

- d. In the **AWS regions** field of the AWS regions and STS assume role name section, enter the AWS regions to collect the CI data.

By default, the Service Graph Connector for AWS runs through all the AWS regions to collect the CI data.

You can enter AWS specific regions to speed up the CI data import process. For example, `us-east1`, `us-east-2`.

If you update the **AWS regions** field value later, clear the value of the **Last run datetime** field in all the data sources

related to the Service Graph Connector for AWS to import a new set of data.

- e. In the **STS assume role name** field of the AWS regions and STS assume role name section, enter the AWS Identity and Access Management (IAM) role name.
The AWS IAM role name is obtained by the ServiceNow user by calling the AssumeRole API offered by the AWS Security Token Service (STS). The AssumeRole API returns a set of temporary security credentials for the ServiceNow user to access the AWS resources.

Note: Enter the IAM role name but don't prefix `arn` in the name. If you leave this field is empty, the value of this field is automatically set to **SnowOrganizationAccountAccessRole**, which is the default IAM role name for the ServiceNow user.

- f. In the SSM SendCommand document details section, enter the name of the document that defines the actions run by the AWS Systems Manager (SSM) on a Linux-based Amazon EC2 instance or a Windows-based Amazon EC2 instance in their respective fields.
- g. In the **Management account ID** field of the Management account ID and standalone account ID section, enter the management account ID in the AWS organization.

Enter a value for this field when the ServiceNow user was created in an AWS member account.

The account calls the ListAccounts API associated with the AWS organization to collect CI information from all the accounts. For more information, see [ListAccounts](#) on the AWS documentation site.

- h. In the **Standalone account ID** field of the Management account ID and standalone account ID section, enter the ID of a member account in the AWS organization.

Note: When specifying a standalone account, the AWS organization-related data such as the organization name, organization units, organization ID, and service accounts are not imported. To import the full data later, clear any value mentioned in the **Standalone account ID** field. See the [Service Graph Connector for AWS - Standalone Setup \[KB1642159\]](#) article in the Now Support Knowledge Base.

- i. In the AWS config aggregator details section, enter the AWS account details for the aggregator resource type.

AWS config aggregator details

Field	Description
Config aggregator account	AWS account where the aggregator resource type in the AWS Config service has been configured. Enter a value in this field when you're using an AWS Config aggregator.
Config aggregator name	Name of the aggregator resource type. This field is available only when you enter a value in the Config aggregator account field.
Config aggregator region	Region where the aggregator resource type resides. This field is available only when you enter a value in the Config aggregator account field.

- j. In the AWS key rotation setup section, enter the key rotation process details.

AWS key rotation setup

Field	Description
AWS rotate keys	Option to enable the key rotation process.
AWS key rotation date	Key rotation date. Set the value to the required key rotation date for the first run. For subsequent runs, this field is automatically set to the rotation date. This field is available only when you select the AWS Rotate Keys check box.
AWS key rotation period (in days)	Key rotation period in days. This field is available only when you select the AWS rotate keys check box.
AWS key rotation status	Status message of a key rotation displaying whether the rotation was a success or a failure. This field is automatically set to display the key rotation status message. This field is available only when you select the AWS rotate keys check box. If the rotation status is a failure, an email notification is triggered, if configured.
Email accounts for receiving error notifications	Comma-separated list of the email addresses of recipients who receive notifications about the AWS key rotation errors.

Field	Description
Email account groups for receiving error notifications	Comma-separated list of the ServiceNow groups who receive notifications about the AWS key rotation errors.

- k. Select the **Is gov cloud** check box in the Gov cloud setup section to indicate that the connection setup is for the AWS GovCloud.
- l. In the SSM EKS SendCommand document details section, enter the AWS SSM document details.

SSM EKS SendCommand document details

Field	Description
EKS cluster names document	Name of the AWS SSM document to discover EKS clusters associated with EC2 Bastion hosts.
EKS shell script document	Name of the AWS SSM document to fetch CIs related to Kubernetes components, such as pods, services, and deployments, from EKS clusters.

- m. Select **Save properties**.
- n. Select **Continue**.
- c. Configure the required EC2 resources for Amazon Elastic Kubernetes Service (EKS) to import EKS cluster data.

Note: Complete this step only when EC2 resources are needed. Else, select **Skip** for the **Configure EKS EC2 resources** activity.

An EKS EC2 resource is a bastion host that has network access to EKS clusters. The EKS clusters aren't directly accessible to the connector. Therefore, you must provide the EKS EC2 resource

details. For importing EKS cluster data, the connector uses the SSM Send Command on EKS EC2 resources to run kubectl commands remotely.

Note: Ensure that you've configured your AWS environment for the EKS integration. For more information, see the [Service Graph Connector for AWS - Amazon EKS Integration \[KB1437138\]](#) article in the Now Support Knowledge Base.

- a. In the **Setup** stage of the playbook, select the **Configure EKS EC2 resources** activity.
- b. On the Configure EKS EC2 resources page, select **New**.
- c. On the Configure EKS EC2 resources window that appears, fill in the fields.

Configure EKS EC2 resources fields

Field	Description
EKS EC2 Resource Id	Identifier of the EKS EC2 resource.
EC2 Region	AWS region where the EKS EC2 resource is located.
EC2 Account	User name assigned to the EKS EC2 resource account.
Connection Alias	Connection alias associated with the AWS environment setup and configured in step 7.a.ii .
Connection	Connection name associated with the AWS environment setup and configured in step 7.a.ii .
Active	Option to activate the EKS EC2 resource.

- d. Select **Save**.

- e. Repeat steps from 7.c.ii to 7.c.iv to add more EKS EC2 resources.
- f. Select **Continue**.
- d. Run the AWS diagnostic tool before running a scheduled import job to identify any issues in the AWS environment setup.
 - a. In the **Setup** stage of the playbook, select the **Run diagnostic tests** activity.
 - b. On the Run diagnostic test page, select an option to exclude the corresponding test results from the diagnostic summary.

Skip SSM setup tests

Excludes the software inventory data from the summary results by not calling the GetInventory API. Select this option when you've opted out or not set up the configuration for SSM.

Skip SSM Deep Discovery tests

Excludes the deep discovery data from the summary results. Select this option when you've opted out or not set up the configuration for SSM deep discovery.

Skip EKS setup tests

Excludes the EKS data from the summary results by not running the kubectl commands.

Note: This check box appears only when you've configured EC2 resources in the **Configure EKS EC2 resources** activity.

- c. Select **Run diagnostic test** and wait for the test to complete.
- d. Review the diagnostic summary, the API access results, and the IAM permission validation logs.
- e. When the test results are successful, select **Continue**.
- e. Configure the import schedule to import data at regular intervals.

- a. In the **Setup** stage of the playbook, select the **Configure import schedule** activity.
- b. Expand the Parent scheduled data import within the Import schedules list to select the **SG-AWS-Organization** import schedule.
- c. In the Configure import schedule dialog box, select the **Active** check box, and then fill in the run schedule and time details.

For more information, see [Schedule a data import](#).

- d. Select **Save**.

Alternatively, select **Execute Now** to execute the import schedule immediately.

- e. Select **Continue**.

- f. In the **Setup** stage of the playbook, select the **Confirm connection setup** activity to verify whether the connection was created.

What to do next

Select **View all connections** to review the connection details. The configured connection appears in the Installed connections list.

Related concepts

- [Service Graph Connector for AWS](#)
- [Accessing the connection details of Service Graph Connector for AWS](#)
- [Additional features within the Service Graph Connector for AWS](#)

Related reference

- [CMDB classes targeted in Service Graph Connector for AWS](#)
- [Supported AWS resource types](#)

Configure Service Graph Connector for AWS using the guided setup

Set up the AWS environment and scheduled jobs to pull in AWS data into the CMDB.

Before you begin

Important: Unless there are configuration issues, use the SGC Central view in the CMDB Workspace to configure the connection for the connector, as the guided setup method is being deprecated.

To use this Service Graph Connector, you need a subscription to a Subscription Unit that is based in the IT Operations Management (ITOM) Visibility application or in the ITOM Discovery application. As defined in the section titled "Managed IT Resource Types" in [ServiceNow Subscription Unit Overview](#) for your subscription, for managed IT resources that are created or modified in the CMDB by this Service Graph Connector, but that aren't yet managed by [ITOM Visibility](#) or [ITOM Discovery](#), these resources will increase Subscription Unit consumption from that application. Review your current Subscription Unit consumption within ITOM Visibility or ITOM Discovery to ensure available capacity.

Dependencies and requirements:

- The [Integration Commons for CMDB](#) store app, which is automatically installed.
- The CMDB CI class models store app, which is automatically installed. See [CMDB CI Class Models](#).
- Discovery Core plugin (com.snc.discovery.core), which is automatically installed by Discovery.
- The ITOM Discovery License plugin (com.snc.itom.discovery.license). You must activate this plugin.
- ITOM Licensing plugin (com.snc.itom.license). For more information, see [Request Discovery](#).

Ensure that you've completed the prerequisites for setting up AWS. See [Configure the AWS environment](#).

Note: When upgrading from an earlier version, ensure that you've updated the SSM documents and IAM permissions in your AWS instance.

- For updating SSM documents, run the SG-AWS-RunShellScript-Setup.yml, SG-AWS-RunPowerShellScript-Setup.yml, and SG-AWS-RunKubeCtlShellScript.yml scripts.
- For updating the IAM permissions assigned to the SnowOrganizationAccountAccessRole role, run the CreateSnowOrganizationAccountAccessRoleInMemberAccount.yml script.

Role required: admin

About this task

For more information on the **Service Graph Connector for AWS** setup instructions, see the following articles:

- [Service Graph Connector for AWS - Setup Instructions \[KB1220597\]](#) article on the Now Support Knowledge Base.
- [Service Graph Connector for AWS - Introduction](#) on the ServiceNow Community site.
- [SGC-AWS - Release 2.0 Features](#) on the ServiceNow Community site.

If you upgrade the connector, complete the following tasks before the upgrade process:

- If there are any customizations, delete the records associated with the Service Graph Connector for AWS from the Customer Updates [sys_update_xml] table. See [Customer Updates table](#).
- Upgrade any dependencies.

After you've installed the latest version of the Service Graph Connector for AWS:

- Perform a full import of data from your AWS instance. Verify that the **Use last run datetime** field value is cleared for all the AWS data sources in the Data Source [sys_data_source] table.
- Download and rerun the AWS scripts. See [Configure the AWS environment](#).

Procedure

1. Ensure that you've selected the **Service Graph Connector for AWS** application scope by using the application picker.
For more information, see [Application picker](#).
2. Navigate to **All > Service Graph Connectors > AWS > Setup**.
3. [Configure the AWS environment](#).

Note: For the Download the scripts task, select **Configure** to download the scripts.

After you've downloaded the scripts and configured the AWS environment, set the Download the scripts and AWS Setup Instructions tasks to complete.

4. Configure the authentication credentials to authenticate requests sent to the AWS APIs.
 - a. Configure your AWS credentials.
 - a. For the Configure the connection section of the Service Graph Connector for AWS page, select **Get Started**.
 - b. For the Configure the credentials task, select **Configure**.
 - c. In the **Name** field, enter a name for the authentication.

SG-AWS-Credentials-0rg is the default credential alias name. You can add multiple AWS instances. However, don't modify the default connection alias.

- d. Enter the access key ID and the secret access key in the **Access Key ID** and **Secret Access Key** fields respectively.

The AWS access keys are long-term credentials for the IAM user and include two parts: an access key ID and a secret

- access key. You must use both the access key ID and the secret access key together to authenticate requests.
- e. Return to the Configure the connection task page by selecting the back icon (<).
 - f. Set the Configure the credentials task to complete by selecting **Mark as Complete**.
- b. Test the AWS API connection to import data from the AWS application.
 - a. For the Test the connection task in the Configure the connection section, select **Configure**.
 - b. Select the **Test Connection** related link.
 - c. When the **Status** field is set to **Success**, select the back icon (<) to return to the guided setup.
 - d. Set the Test the connection task to complete by selecting **Mark as Complete**.
 - c. Update the required configuration properties for an AWS instance.
 - a. For the Update configuration properties for instance task, select **Configure**.
 - b. On the SG-AWS Configuration Properties form that opens in a new tab, review, and modify the fields.

SG-AWS Configuration Properties form

Field	Description
Connection Details	
Connection Alias	Name to identify the AWS connection record. For example, SG_AWS_CredentialAlias_Org. You can add multiple AWS instances. However, don't

Field	Description
	modify the name for the default connection alias SG_AWS_CredentialAlias_Org.
Organization Details	
Organization Account	Numeric account identifier of the AWS organization.
Organization Name	Name of the AWS organization.
Organization Description	Description of the AWS organization.
AWS Regions	
Regions	AWS regions to collect the CI data. By default, the Service Graph Connector for AWS runs through all the AWS regions to collect the CI data. You can enter AWS specific regions to speed up the CI data import process. For example, us-east1, us-east-2. If this field is left empty, the Service Graph Connector for AWS pulls the resources from all the AWS regions. However, for the AWS GovCloud regions, don't leave the Regions field

Field	Description
	empty. The supported AWS GovCloud regions are <code>us-gov-east-1</code> and <code>us-gov-west-1</code>. If you update the Regions field value later, clear the value of the Last run datetime field in all the data sources related to the Service Graph Connector for AWS to import a new set of data.
STS Assume Role Name	
STS Role	AWS Identity and Access Management (IAM) role name that is obtained by the ServiceNow user by calling the AssumeRole API offered by the AWS Security Token Service (STS). The AssumeRole API returns a set of temporary security credentials for the ServiceNow user to access the AWS resources. Note: Enter the IAM role name but don't prefix <code>arn</code> in the name. If you leave this field is empty, the value of this field is automatically set to SnowOrganizationAccountAccessRole, which is the default IAM role name for the ServiceNow user.

Field	Description
S3 Account Details	
S3 Account Id	Numeric identifier of the AWS account that hosts the Amazon Simple Storage Service (Amazon S3) bucket.
S3 Bucket Name	Name of the Amazon S3 bucket that collects the details from Amazon EC2 instances.
S3 Region	Region where the Amazon S3 bucket resides.
SSM SendCommand Document Details	
SSM Send Command Linux Name	Name of the document that defines the actions run by the AWS Systems Manager (SSM) on a Linux-based Amazon EC2 instance.
SSM Send Command Windows Name	Name of the document that defines the actions run by the AWS SSM on a Windows-based Amazon EC2 instance.
Management Account ID	
Management Account ID	Management account in the AWS organization. The account calls the ListAccounts API associated with the AWS organization to collect CI information from all the accounts. For more information, see ListAccounts on the AWS documentation site.

Field	Description
	Enter a value for this field when the ServiceNow user was created in an AWS member account.
Standalone Account ID Details	
Standalone Account ID	ID of a member account in the AWS organization. Note: When specifying a standalone account, AWS organization-related data such as the organization name, organization units, organization ID, and service accounts are not imported. If you want to import the full data later, clear any value mentioned in the Standalone Account ID field. See the Service Graph Connector for AWS - Standalone Setup [KB1642159] article in the Now Support Knowledge Base.
AWS Config Aggregator Details	
Config Aggregator Account	AWS account where the aggregator resource type in the AWS Config service has been configured.

Field	Description
	Enter a value in this field when you're using an AWS Config aggregator.
Config Aggregator Name	Name of the aggregator resource type. This field is available only when you enter a value in the Config Aggregator Account field.
Config Aggregator Region	Region where the aggregator resource type resides. This field is available only when you enter a value in the Config Aggregator Account field.
AWS Key Rotation Setup	
AWS Rotate Keys	Option to enable the key rotation process.
AWS Key Rotation Date	Key rotation date. Set the value to the required key rotation date for the first run. For subsequent runs, this field is automatically set to the rotation date. This field is available only when you select the AWS Rotate Keys check box.
AWS Key Rotation Period (in Days)	Key rotation period in days. This field is available only when you select the AWS Rotate Keys check box.
AWS Key Rotation Status	Status message of a key rotation displaying whether the rotation was a success

Field	Description
	or a failure. This field is automatically set to display the key rotation status message. This field is available only when you select the AWS Rotate Keys check box. If the rotation status is a failure, an email notification is triggered, if configured.
Email accounts for receiving error notifications	Comma-separated list of the email addresses of recipients who receive notifications about the AWS key rotation errors.
Email account groups for receiving error notifications	Comma-separated list of the ServiceNow groups who receive notifications about the AWS key rotation errors.
Gov Cloud Setup	
Is Gov Cloud	Option to indicate that the connection setup is for the AWS GovCloud.
SSM EKS SendCommand Document Details	
EKS Cluster Names Document	Name of the AWS SSM document to discover EKS clusters associated with EC2 Bastion hosts.
EKS Shell Script Document	Name of the AWS SSM document to fetch CIs related to Kubernetes components, such as pods, services, and deployments, from EKS clusters.

- c. Select **Save** and then **Close** to close the tab and return to the guided setup tab.
 - d. Set the Update configuration properties for instance task to complete by selecting **Mark as Complete**.
5. Configure the required EC2 resources for Amazon Elastic Kubernetes Service (EKS) to import EKS cluster data.
An EKS EC2 resource is a bastion host that has network access to EKS clusters. The EKS clusters aren't directly accessible to the connector. Therefore, you must provide the EKS EC2 resource details. For importing EKS cluster data, the connector uses the SSM Send Command on EKS EC2 resources to run kubectl commands remotely.

Note: Ensure that you've configured your AWS environment for the EKS integration. For more information, see the [Service Graph Connector for AWS - Amazon EKS Integration \[KB1437138\]](#) article in the Now Support Knowledge Base.

- a. For the Configure the EKS Resource Details section of the Service Graph Connector for AWS page, select **Get Started**.
 - b. For the Enter the EKS EC2 Resource Details task, select **Configure**.
 - c. On the form that opens in a new tab, fill in the fields.

SG-AWS-EKS-EC2-Resource form

Field	Description
EKS EC2 Resource Id	Identifier of the EKS EC2 resource.
EC2 Account	User name assigned to the EKS EC2 resource account.
EC2 Region	AWS region where the EKS EC2 resource is located.
Active	Option to activate the EKS EC2 resource.

Field	Description
	Note: Set to false, if you are not using the EKS EC2 resource resource.
Connection	Connection alias associated with the AWS environment setup and configured in step 4.a .

- d. Select **Submit** to return to the guided setup.
 - e. Repeat steps from [5.b](#) to [5.d](#) to add multiple EKS EC2 resources. All the EKS EC2 resources are added to the SG-AWS-EKS-Master [sn_aws_integ_sg_aws_eks_master] table.
 - f. Set the Enter the EKS EC2 Resource Details task to complete by selecting **Mark as Complete**.
6. Run the AWS diagnostic tool before running a scheduled import job to identify any issues in the AWS environment setup.
 - a. For the Service Graph AWS Diagnostic Tool section of the Service Graph Connector for AWS page, select **Get Started**.
 - b. For the AWS Setup Diagnostic Tool task, select **Configure**.
 - c. Select the organization ID from the text field.
 - d. Select **Run Diagnostic Test**.

Tip: Select one of the following options to exclude the corresponding test results from the diagnostic summary:

Skip SSM setup tests

Excludes the software inventory data from the summary results by not calling the GetInventory API. Select this option when you've opted out or not set up the configuration for SSM.

Skip SSM Deep Discovery tests

Excludes the deep discovery data from the summary results. Select this option when you've opted out or not set up the configuration for SSM deep discovery.

Skip EKS setup tests

Excludes the EKS data from the summary results by not running the kubectl commands. Select this option when you've opted out or not set up the EKS integration.

- e. (Optional) View only EKS cluster test results by selecting **View EKS Test Details**.
 - f. (Optional) Preview any previous diagnostic tool results by selecting **Load DT Results**, selecting a diagnostic ID, and then selecting **Load Results**.
 - g. When you finish reviewing the diagnostic summary results, select the back button of your browser to return to the guided setup.
 - h. Set the AWS Setup Diagnostic Tool task to complete by selecting **Mark as Complete**.
7. Configure the scheduled jobs to import data from the AWS application.
 - a. In the Configure the scheduled import jobs section of the Service Graph Connector for AWS page, select **Get started**.
 - b. For the Configure the scheduled job task, select **Configure**.
 - c. Select the scheduled job that you want to activate.

- d. On the Scheduled Data Import form, verify the field values for the scheduled job.
For more information, see [Schedule a data import](#).
 - e. Select **Update**.
 - f. Repeat the steps 7.c to 7.e for each scheduled job for data import.
 - g. Select the back icon (<) to return to the guided setup page.
 - h. Set the Configure the scheduled job task to complete by selecting **Mark as Complete** in the guided setup.
8. (Optional) Add multiple AWS instances.
 - a. In the Add Multiple Instances section of the Service Graph Connector for AWS page, select **Get Started**.
 - b. Ensure that you have edit permissions for the Datasource [sys_data_source] table to create data sources for the new instance.
 - a. Select the **Global** application scope by using the application picker.
 - b. For the Update Data Source Access task, select **Configure**.
 - c. In the Application Access related list, select the **Can create**, **Can update**, and **Can delete** check boxes, if not already selected.
 - d. Select **Update** to close the tab and return to the guided setup.
 - e. Modify the application scope again to **Service Graph Connector for AWS** by using the application picker.
 - f. Set the Update Data Source Access task to complete by selecting **Mark as Complete**.
 - c. Update access for scheduled import jobs.
 - a. Select the **Global** application scope by using the application picker.

- b. For the Update Scheduled Data Import Access task, select **Configure**.
 - c. In the Application Access related list, select the **Can create**, **Can update**, and **Can delete** check boxes, if not already selected.
 - d. Select **Update** to close the tab and return to the guided setup tab.
 - e. Set the Update Scheduled Data Import Access task to complete by selecting **Mark as Complete**.
 - f. Modify the application scope again to **Service Graph Connector for AWS** by using the application picker.
- d. Clear the cache of the Data Source [sys_data_source] and Scheduled Data Imports [scheduled_import_set] tables to enable creating a data source for the new connection in the Service Graph Connector for AWS.
 - a. For the Clear Cache for Data Source and Scheduled Data Imports tables task, select **Configure**.
 - b. In the **Run script** text box of the background script page, enter the following script:

```
GlideTableManager.invalidateTable("sys_data_source");
    GlideCacheManager.flushTable("sys_data_source");

    GlideTableManager.invalidateTable("scheduled_import_set");
    GlideCacheManager.flushTable("scheduled_import_set");

    GlideTableManager.invalidateTable("sys_db_object");
    GlideCacheManager.flushTable("sys_db_object");
```

- c. Select **Run Script** to run the background script in the **global** scope.

The script may take several minutes to execute.

- d. After the script is executed, select **Close**.
 - e. Set the Clear Cache for Data Source and Scheduled Data Imports tables task to complete by selecting **Mark as Complete**.
 - f. Modify the application scope again to **Service Graph Connector for AWS** by using the application picker.
- e. Create a credential alias for the new AWS connection in the Service Graph Connector for AWS.
- a. For the Create new Connection & Credentials Alias Record task, select **Configure**.
 - b. On the Connection & Credential Aliases form that opens in a new tab, fill in the connection details.
 - c. Select **Submit** and close the tab and return to the guided setup tab.
 - d. Set the Create new Connection & Credentials Alias Record task to complete by selecting **Mark as Complete**.
- f. Create credentials for the new AWS credential alias.
- a. For the Create new Connection & Credentials Alias Record task, select **Configure**.
 - b. On the Connections page of the Workflow Studio, select **Add Connection**.
 - c. In the Create Connection window, fill in the connection name, access key, and secret key details.
 - d. Select **Create Connection**.
 - e. Close the Workflow Studio and return to the guided setup tab.
 - f. Set the Create new Connection & Credentials Alias Record task to complete by selecting **Mark as Complete**.

- g. Update properties for the new instance for the Service Graph Connector for AWS by selecting **Configure** for the Configure AWS environment for the new Instance task.

For more information, follow the step 4.c discussed earlier for configuring the properties of the AWS connection available by default.

When you're done updating the properties, set the Configure AWS environment for the new Instance task to complete by selecting **Mark as Complete**.

- h. Enter the EKS EC2 Resource details by selecting **Configure** for the Update EKS Resource Details task.

For more information, follow the step 5 discussed earlier for configuring the EKS EC2 Resource details for the AWS connection available by default.

When you're done updating the properties, set the Update EKS Resource Details task to complete by selecting **Mark as Complete**.

- i. Configure the scheduled imports for the new AWS instance.

- a. For the Configure the Scheduled Imports task, select **Configure**.

- b. In the Scheduled Data Imports list that opens in a new tab, select the organization of the AWS instance that you want to configure.

- c. Select the scheduled data import that you want to activate.

- d. On the Scheduled Data Import form, modify the field values for the scheduled job.

- e. Select **Update**.

- f. Repeat the steps 8.i.iii to 8.i.v for each scheduled job for data import.

- g. Close the tab for the Scheduled Data Imports list and return to the guided setup tab.

- h. Set the Configure the Scheduled Imports task to complete by selecting **Mark as Complete** in the guided setup.

Related concepts

- [Handling import failures for large AWS resources](#)
- [Handling memory utilization errors for Service Graph Connector for AWS](#)

CMDB classes targeted in Service Graph Connector for AWS

When you complete setting up the connection, you can configure the integration to periodically pull data from AWS. The data is saved in tables that extend from the Configuration item [cmdb_ci] table.

Amazon Redshift [cmdb_ci_aws_redshift]

The following attributes in the Amazon Redshift [cmdb_ci_aws_redshift] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
TCP port(s)	tcp_port
Cluster Availability Status	cluster_availability_status
Install Status	install_status
Availability Zone	availability_zone
Fully qualified domain name	fqdn
Node Count	node_count
Node Type	node_type
Operational status	operational_status

Attribute label	Attribute name
Start date	start_date
VPC ID	vpc_id

Relationships created for Application

Parent class	Relationship type	Child class
Amazon Redshift [cmdb_ci_aws_redshift]	Hosted on::Hosts	AWS Datacenter [cmdb_ci_aws_datacenter]
Amazon Redshift [cmdb_ci_aws_redshift]	Reference	Availability Zone [cmdb_ci_availability_zone]

Application [cmdb_ci_appl]

The following attributes in the Application [cmdb_ci_appl] table are populated by collected data:

Attribute label	Attribute name
Class	sys_class_name
Version	version
TCP port(s)	tcp_port
Running Process	running_process
Running process key parameters	running_process_key_parameters
Name	name
Running process command	running_process_command

Relationship created for Application

Parent class	Relationship type	Child class
Application [cmdb_ci_appl]	Runs on::Runs	Server [cmdb_ci_server]

Availability Zone [cmdb_ci_availability_zone]

The following attributes in the Availability Zone [cmdb_ci_availability_zone] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Install Status	install_status
Operational status	operational_status
State	state

Relationships created for Availability Zone

Parent class	Relationship type	Child class
Availability Zone [cmdb_ci_availability_zone]	Contains::Contained by	Cloud Subnet [cmdb_ci_cloud_subnet]
Availability Zone [cmdb_ci_availability_zone]	Contains::Contained by	Cloud Load Balancer [cmdb_ci_cloud_load_balancer]

AWS Datacenter [cmdb_ci_aws_datacenter]

The following attributes in the AWS Datacenter [cmdb_ci_aws_datacenter] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Region	region
Install Status	install_status
Operational status	operational_status

Relationships created for AWS Datacenter

Parent class	Relationship type	Child class
AWS Datacenter [cmdb_ci_aws_datacenter]	Contains::Contained by	Availability Zone [cmdb_ci_availability_zone]
AWS Datacenter [cmdb_ci_aws_datacenter]	Hosted on::Hosts	Cloud Service Account [cmdb_ci_cloud_service_account]

AWS Organizational Unit [cmdb_ci_aws_org_unit]

The following attributes in the AWS Organizational Unit [cmdb_ci_aws_org_unit] table are populated by collected data:

Attribute label	Attribute name
Object ID	object_id
Install Status	install_status
Name	name
Operational status	operational_status
Org Unit Parent ID	org_unit_parent_id

Attribute label	Attribute name
Organizational ID	aws_org_id
Root ID	root_id

Relationship created for AWS Organizational Unit

Parent class	Relationship type	Child class
AWS Organizational Unit [cmdb_ci_aws_org_unit]	Contains::Contained by	Cloud Service Account [cmdb_ci_cloud_service_account]

Block Endpoint [cmdb_ci_endpoint_block]

The following attributes in the Block Endpoint [cmdb_ci_endpoint_block] table are populated by collected data:

Attribute label	Attribute name
Host	host
Name	name
Object ID	object_id
Install Status	install_status
Operational status	operational_status

Cloud DataBase [cmdb_ci_cloud_database]

The following attributes in the Cloud DataBase [cmdb_ci_cloud_database] table are populated by collected data:

Attribute label	Attribute name
Name	name

Attribute label	Attribute name
Object ID	object_id
TCP port(s)	tcp_port
Automated Backups	automated_backup
Category	category
Fully qualified domain name	fqdn
Install Status	install_status
Operational status	operational_status
State	state
Type	type
Version	version

Relationships created for Cloud DataBase

Parent class	Relationship type	Child class
Cloud DataBase [cmdb_ci_cloud_data base]	Hosted on::Hosts	AWS Datacenter [cmdb_ci_aws_datac enter]
Cloud DataBase [cmdb_ci_cloud_data base]	Reference	Key Value [cmdb_key_value]

Cloud Function [cmdb_ci_cloud_function]

The following attributes in the Cloud Function [cmdb_ci_cloud_function] table are populated by collected data:

Attribute label	Attribute name
Name	name

Attribute label	Attribute name
Object ID	object_id
Code Size	code_size
CodeSha256	codesha256
Function Last Modified	function_last_modified
Install Status	install_status
Language	language
Operational status	operational_status
Version	version

Relationships created for Cloud Function

Parent class	Relationship type	Child class
Cloud Function [cmdb_ci_cloud_function]	Hosted on::Hosts	AWS Datacenter [cmdb_ci_aws_datacenter]
Cloud Function [cmdb_ci_cloud_function]	Reference	Key Value [cmdb_key_value]

Cloud Gateway [cmdb_ci_cloud_gateway]

The following attributes in the Cloud Gateway [cmdb_ci_cloud_gateway] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Fully qualified domain name	fqdn

Attribute label	Attribute name
Install Status	install_status
Operational status	operational_status

Relationships created for Cloud Gateway

Parent class	Relationship type	Child class
Cloud Gateway [cmdb_ci_cloud_gateway]	Hosted on::Hosts	AWS Datacenter [cmdb_ci_aws_datacenter]
Cloud Gateway [cmdb_ci_cloud_gateway]	Reference	Key Value [cmdb_key_value]

Cloud Hardware Type [cmdb_ci_cloud_hardware_type]

The following attributes in the Cloud Hardware Type [cmdb_ci_cloud_hardware_type] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Provider	provider
Cores	cores
Install Status	install_status
IP Address	ip_address
Local Storage GB	local_storage_gb
Memory MB	memory_mb

Attribute label	Attribute name
Operational status	operational_status
vCPUs	vcpus
Zone	zone

Relationships created for Cloud Hardware Type

Parent class	Relationship type	Child class
Cloud Hardware Type [cmdb_ci_cloud_hardware_type]	Hosted on::Hosts	AWS Datacenter [cmdb_ci_aws_datacenter]

Cloud Image [cmdb_ci_cloud_os_image]

The following attributes in the Cloud Image [cmdb_ci_cloud_os_image] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Description	short_description
Environment	environment
Guest OS	guest_os
Image Source	image_source
Image Type	image_type
Provider	provider
Root Device Type	root_device_type

Relationships created for Cloud Image

Parent class	Relationship type	Child class
Cloud Image [cmdb_ci_cloud_os_image]	Hosted on::Hosts	AWS Datacenter [cmdb_ci_aws_datacenter]
Cloud Image [cmdb_ci_cloud_os_image]	Reference	Key Value

Cloud Load Balancer [cmdb_ci_cloud_load_balancer]

The following attributes in the Cloud Load Balancer [cmdb_ci_cloud_load_balancer] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Canonical Hosted Zone ID	canonical_hosted_zone_id
Canonical Hosted Zone Name	canonical_hosted_zone_name
DNS Name	dns_name
Fully qualified domain name	fqdn
Install Status	install_status
Operational status	operational_status
State	state

Relationships created for Cloud Load Balancer

Parent class	Relationship type	Child class
Cloud Load Balancer [cmdb_ci_cloud_load_balancer]	Contains::Contained by	Compute Security Group [cmdb_ci_compute_security_group]
Cloud Load Balancer [cmdb_ci_cloud_load_balancer]	Hosted on::Hosts	AWS Datacenter [cmdb_ci_aws_datacenter]
Cloud Load Balancer [cmdb_ci_cloud_load_balancer]	Reference	Key Value [cmdb_key_value]

Cloud Mgmt Network Interface [cmdb_ci_nic]

The following attributes in the Cloud Mgmt Network Interface [cmdb_ci_nic] table are populated by collected data:

Attribute label	Attribute name
IP Address	ip_address
Name	name
Object ID	object_id
Public IP	public_ip
Install Status	install_status
Operational status	operational_status
Private DNS	private_dns
Private IP	private_ip
Public DNS	public_dns

Attribute label	Attribute name
State	state
Configuration Item	cmdb_ci

Relationships created for Cloud Mgmt Network Interface

Parent class	Relationship type	Child class
Cloud Mgmt Network Interface [cmdb_ci_nic]	Hosted on::Hosts	AWS Datacenter [cmdb_ci_aws_datacenter]
Cloud Mgmt Network Interface [cmdb_ci_nic]	Use End Point To::Use End Point From	VNIC Endpoint [cmdb_ci_endpoint_vnic]
Cloud Mgmt Network Interface [cmdb_ci_nic]	Reference	Key Value [cmdb_key_value]
Cloud Mgmt Network Interface [cmdb_ci_nic]	Reference	Virtual Machine Instance [cmdb_ci_vm_instance]

Cloud Network [cmdb_ci_network]

The following attributes in the Cloud Network [cmdb_ci_network] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Cidr	cidr
Install Status	install_status

Attribute label	Attribute name
Operational status	operational_status
State	state

Relationships created for Cloud Network

Parent class	Relationship type	Child class
Cloud Network [cmdb_ci_network]	Hosted on::Hosts	AWS Datacenter [cmdb_ci_aws_datacenter]
Cloud Network [cmdb_ci_network]	Contains::Contained by	Cloud Subnet [cmdb_ci_cloud_subnet]
Cloud Network [cmdb_ci_network]	Contains::Contained by	Compute Security Group [cmdb_ci_compute_security_group]
Cloud Network [cmdb_ci_network]	Reference	Key Value [cmdb_key_value]

Cloud Object Storage [cmdb_ci_cloud_object_storage]

The following attributes in the Cloud Object Storage [cmdb_ci_cloud_object_storage] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
ACL Access Type	acl_access_type
Cloud Provider	cloud_provider

Attribute label	Attribute name
Creation Date	creation_date
Encryption Type	encryption_type
Install Status	install_status
Operational status	operational_status
Policy Access Type	policy_access_type
Service Name	service_name

Relationships created for Cloud Object Storage

Parent class	Relationship type	Child class
Cloud Object Storage [cmdb_ci_cloud_object_storage]	Hosted on::Hosts	AWS Datacenter [cmdb_ci_aws_datacenter]
Cloud Object Storage [cmdb_ci_cloud_object_storage]	Reference	Key Value [cmdb_key_value]

Cloud Organizations [cmdb_ci_cloud_org]

The following attributes in the Cloud Organizations [cmdb_ci_cloud_org] table are populated by collected data:

Attribute label	Attribute name
Object ID	object_id
Install Status	install_status
Master Email	master_email
Name	name
Operational status	operational_status

Attribute label	Attribute name
Root ID	root_id

Relationships created for Cloud Organizations

Parent class	Relationship type	Child class
Cloud Organizations [cmdb_ci_cloud_org]	Contains::Contained by	Cloud Service Account [cmdb_ci_cloud_service_account]
Cloud Organizations [cmdb_ci_cloud_org]	Contains::Contained by	AWS Organizational Unit [cmdb_ci_aws_org_unit]

Cloud Resource [cmdb_ci_cmp_resource]

The following attributes in the Cloud Resource [cmdb_ci_cmp_resource] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Description	short_description
Install Status	install_status
Operational status	operational_status
Resource type	resource_type
State	state

Relationship created for Cloud Resource

Parent class	Relationship type	Child class
Cloud Resource [cmdb_ci_cmp_resource]	Reference	Key Value [cmdb_key_value]

Cloud Service Account [cmdb_ci_cloud_service_account]

The following attributes in the Cloud Service Account [cmdb_ci_cloud_service_account] table are populated by collected data:

Attribute label	Attribute name
Name	name
Account Email	account_email
Install Status	install_status
Is management account	is_master_account
Operational status	operational_status
Parent account	parent_account
Datacenter Type	datacenter_type
Discovery credentials	discovery_credentials
Account Id	account_id
Object ID	object_id
Organization Id	organization_id

Relationships created for Cloud Service Account

Parent class	Relationship type	Child class
Cloud Service Account [cmdb_ci_cloud_service_account]	Reference	Cloud Service Account [cmdb_ci_cloud_service_account]
Cloud Service Account [cmdb_ci_cloud_service_account]	Reference	Key Value [cmdb_key_value]

Cloud Subnet [cmdb_ci_cloud_subnet]

The following attributes in the Cloud Subnet [cmdb_ci_cloud_subnet] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Available IP Count	available_ip_count
CIDR	cidr
Install Status	install_status
Operational status	operational_status
State	state

Relationships created for Cloud Subnet

Parent class	Relationship type	Child class
Cloud Subnet [cmdb_ci_cloud_subnet]	Contains::Contained by	Cloud Load Balancer [cmdb_ci_cloud_load_balancer]

Parent class	Relationship type	Child class
Cloud Subnet [cmdb_ci_cloud_subnet]	Contains::Contained by	Cloud Mgmt Network Interface [cmdb_ci_nic]
Cloud Subnet [cmdb_ci_cloud_subnet]	Reference	Key Value [cmdb_key_value]

Compute Security Group [cmdb_ci_compute_security_group]

The following attributes in the Compute Security Group [cmdb_ci_compute_security_group] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Install Status	install_status

Relationships created for Compute Security Group Table

Parent class	Relationship type	Child class
Compute Security Group [cmdb_ci_compute_security_group]	Hosted on::Hosts	AWS Datacenter [cmdb_ci_aws_datacenter]
Compute Security Group [cmdb_ci_compute_security_group]	Reference	Key Value [cmdb_key_value]

Docker Container [cmdb_ci_docker_container]

The following attributes in the Docker Container [cmdb_ci_docker_container] table are populated by collected data:

Attribute label	Attribute name
Container id	container_id
Image id	image_id
Install Status	install_status
Name	name
Operational status	operational_status
Status	status

Docker Image [cmdb_ci_docker_image]

The following attributes in the Docker Image [cmdb_ci_docker_image] table are populated by collected data:

Attribute label	Attribute name
Image id	image_id
Install Status	install_status
Name	name
Operational status	operational_status

Relationship created for Docker Image

Parent class	Relationship type	Child class
Docker Image [cmdb_ci_docker_image]	Instantiates::Instantiated by	Docker Container [cmdb_ci_docker_container]

DynamoDB Table [cmdb_ci_dynamodb_table]

The following attributes in the DynamoDB Table [cmdb_ci_dynamodb_table] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Cloud Provider	cloud_provider
Creation Date	creation_date
Encryption	encryption
Install Status	install_status
Operational status	operational_status
Read Units	read_units
Service Name	service_name
Write Units	write_units

Relationships created for DynamoDB Table

Parent class	Relationship type	Child class
DynamoDB Table [cmdb_ci_dynamodb_table]	Hosted on::Hosts	AWS Datacenter [cmdb_ci_aws_datacenter]
DynamoDB Table [cmdb_ci_dynamodb_table]	Reference	Key Value [cmdb_key_value]

Hardware Type [cmdb_ci_compute_template]

The following attributes in the Hardware Type [cmdb_ci_compute_template] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Cores	cores
Install Status	install_status
Local Storage GB	local_storage_gb
Memory MB	memory_mb
Operational status	operational_status
vCPUs	vcpus

Note: If the `sn_itom_pattern.use a single hardware type` for cloud data center property is set to `true`, hardware type data is populated in the Cloud Hardware Type `[cmdb_ci_cloud_hardware_type]` table. If this property is set to `false`, hardware type data is populated in the Hardware Type `[cmdb_ci_compute_template]` table. Before you configure the `sn_itom_pattern.use a single hardware type` for cloud data center property, the existing hardware type data must be cleaned. For instructions, see the [Service Graph Connector For AWS - Migrating to a new hardware type model \[KB1705233\]](#) article in the Now Support Knowledge Base.

Relationship created for Hardware Type

Parent class	Relationship type	Child class
Hardware Type <code>[cmdb_ci_compute_template]</code>	Hosted on::Hosts	AWS Datacenter <code>[cmdb_ci_aws_datacenter]</code>

Image [cmdb_ci_os_template]

The following attributes in the Image [cmdb_ci_os_template] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Class	sys_class_name
Description	short_description
Environment	environment
Guest OS	guest_os
Image Source	image_source
Image Type	image_type
Install Status	install_status
Operational status	operational_status
Root Device Type	root_device_type

Relationships created for Image

Parent class	Relationship type	Child class
Image [cmdb_ci_os_template]	Hosted on::Hosts	AWS Datacenter [cmdb_ci_aws_datacenter]
Image [cmdb_ci_os_template]	Reference	Key Value

IP Address [cmdb_ci_ip_address]

The following attributes in the IP Address [cmdb_ci_ip_address] table are populated by collected data:

Attribute label	Attribute name
Install Status	install_status
Nic	nic
IP Address	ip_address
Name	name
Operational status	operational_status

Relationship created for IP Address

Parent class	Relationship type	Child class
IP Address [cmdb_ci_ip_address]	Reference	Network Adapter [cmdb_ci_network_adapter]

Key Value [cmdb_key_value]

The following attributes in the Key Value [cmdb_key_value] table are populated by collected data:

Attribute label	Attribute name
Key	key
Value	Value

Kubernetes Cluster [cmdb_ci_kubernetes_cluster]

The following attributes in the Kubernetes Cluster [cmdb_ci_kubernetes_cluster] table are populated by collected data:

Attribute label	Attribute name
Cluster Resource ID	cluster_resource_id
IP Address	ip_address
Name	name
Port	port
Fully qualified domain name	fqdn
Install status	install_status
Namespace	namespace
Operational status	operational_status

Relationships created for Kubernetes Cluster

Parent class	Relationship type	Child class
Kubernetes Cluster [cmdb_ci_kubernetes_cluster]	Contains::Contained by	Kubernetes Namespace [cmdb_ci_kubernetes_namespace]
Kubernetes Cluster [cmdb_ci_kubernetes_cluster]	Contains::Contained by	Kubernetes Pod [cmdb_ci_kubernetes_pod]
Kubernetes Cluster [cmdb_ci_kubernetes_cluster]	Contains::Contained by	Kubernetes Service [cmdb_ci_kubernetes_service]
Kubernetes Cluster [cmdb_ci_kubernetes_cluster]	Cluster of::Cluster	Kubernetes Node [cmdb_ci_kubernetes_node]

Parent class	Relationship type	Child class
Kubernetes Cluster [cmdb_ci_kubernetes_cluster]	Hosted on::Hosts	AWS Datacenter [cmdb_ci_aws_datacenter]
Kubernetes Cluster [cmdb_ci_kubernetes_cluster]	Managed by::Manages	Server [cmdb_ci_server]
Kubernetes Cluster [cmdb_ci_kubernetes_cluster]	Reference	Key Value [cmdb_key_value]

Kubernetes DaemonSet [cmdb_ci_kubernetes_daemonset]

The following attributes in the Kubernetes DaemonSet [cmdb_ci_kubernetes_daemonset] table are populated by collected data:

Attribute label	Attribute name
Kubernetes UID	k8s_uid
Name	name
Namespace	namespace
Install Status	install_status
Operational status	operational_status
SelfLink	self_link

Relationship created for Kubernetes DaemonSet

Parent class	Relationship type	Child class
Kubernetes DaemonSet [cmdb_ci_kubernetes_daemonset]	Hosted on::Hosts	Kubernetes Cluster [cmdb_ci_kubernetes_cluster]

Kubernetes Deployment [cmdb_ci_kubernetes_deployment]

The following attributes in the Kubernetes Deployment [cmdb_ci_kubernetes_deployment] table are populated by collected data:

Attribute label	Attribute name
Kubernetes UID	k8s_uid
Name	name
Namespace	namespace
Available Replicas	available_replicas
Install Status	install_status
Operational status	operational_status
SelfLink	self_link

Relationship created for Kubernetes Deployment

Parent class	Relationship type	Child class
Kubernetes Deployment [cmdb_ci_kubernetes_deployment]	Hosted on::Hosts	Kubernetes Cluster [cmdb_ci_kubernetes_cluster]

Kubernetes Namespace [cmdb_ci_kubernetes_namespace]

The following attributes in the Kubernetes Namespace [cmdb_ci_kubernetes_namespace] table are populated by collected data:

Attribute label	Attribute name
Kubernetes UID	k8s_uid
Name	name

Attribute label	Attribute name
Install Status	install_status
Namespace	namespace
Operational status	operational_status

Kubernetes Node [cmdb_ci_kubernetes_node]

The following attributes in the Kubernetes Node [cmdb_ci_kubernetes_node] table are populated by collected data:

Attribute label	Attribute name
Kubernetes UID	k8s_uid
Name	name
Install Status	install_status
Operational status	operational_status

Kubernetes Pod [cmdb_ci_kubernetes_pod]

The following attributes in the Kubernetes Pod [cmdb_ci_kubernetes_pod] table are populated by collected data:

Attribute label	Attribute name
Kubernetes UID	k8s_uid
Name	name
Namespace	namespace
Install Status	install_status
IP Address	ip_address
Operational status	operational_status

Attribute label	Attribute name
Resource version	resource_version

Relationships created for Kubernetes Pod

Parent class	Relationship type	Child class
Kubernetes Pod [cmdb_ci_kubernetes_pod]	Contains::Contained by	Docker Container [cmdb_ci_docker_container]
Kubernetes Pod [cmdb_ci_kubernetes_pod]	Contains::Contained by	Kubernetes Volume [cmdb_ci_kubernetes_volume]
Kubernetes Pod [cmdb_ci_kubernetes_pod]	Contains::Contained by	Docker Image [cmdb_ci_docker_image]

Kubernetes ReplicaSet [cmdb_ci_kubernetes_replicaset]

The following attributes in the Kubernetes ReplicaSet [cmdb_ci_kubernetes_replicaset] table are populated by collected data:

Attribute label	Attribute name
Kubernetes UID	k8s_uid
Name	name
Namespace	namespace
Install Status	install_status
Operational status	operational_status
SelfLink	self_link

Relationships created for Kubernetes ReplicaSet

Parent class	Relationship type	Child class
Kubernetes ReplicaSet [cmdb_ci_kubernetes_replicaset]	Hosted on::Hosts	Kubernetes Cluster [cmdb_ci_kubernetes_cluster]

Kubernetes Service [cmdb_ci_kubernetes_service]

The following attributes in the Kubernetes Service [cmdb_ci_kubernetes_service] table are populated by collected data:

Attribute label	Attribute name
Kubernetes UID	k8s_uid
Name	name
Namespace	namespace
Install Status	install_status
IP Address	ip_address
Operational status	operational_status
Selector	selector

Kubernetes Volume [cmdb_ci_kubernetes_volume]

The following attributes in the Kubernetes Volume [cmdb_ci_kubernetes_volume] table are populated by collected data:

Attribute label	Attribute name
Kubernetes UID	k8s_uid
Mount Path	mount_path
Name	name

Attribute label	Attribute name
Volume ID	volume_id
Install Status	install_status
Namespace	namespace
Operational status	operational_status

Network Adapter [cmdb_ci_network_adapter]

The following attributes in the Network Adapter [cmdb_ci_network_adapter] table are populated by collected data:

Attribute label	Attribute name
IP Address	ip_address
MAC Address	mac_address
Configuration Item	cmdb_ci
Name	name
Install Status	install_status
Operational status	operational_status

Relationship created for Network Adapter

Parent class	Relationship type	Child class
Network Adapter [cmdb_ci_network_adapter]	Reference	Server [cmdb_ci_server], Linux Server [cmdb_ci_linux_server], or Windows Server [cmdb_ci_win_server]

Server [cmdb_ci_server]

The following attributes in the Server [cmdb_ci_server] table are populated by collected data:

Attribute label	Attribute name	Data source
Name	name	SG-AWS-SSM-GetS3Object SG-AWS-Get-Inventory (if the AWS SSM service is enabled) SG-AWS-EC2 (if the AWS SSM service isn't enabled)
Class	sys_class_name	SG-AWS-Software-Inventory (if the AWS SSM service is enabled)
CPU core count	cpu_core_count	SG-AWS-SSM-GetS3Object
CPU core thread	cpu_core_thread	SG-AWS-SSM-GetS3Object
CPU count	cpu_count	SG-AWS-SSM-GetS3Object

Attribute label	Attribute name	Data source
CPU name	cpu_name	SG-AWS-SSM-GetS3Object
CPU manufacturer	cpu_manufacturer	SG-AWS-SSM-GetS3Object
CPU speed (MHz)	cpu_speed	SG-AWS-SSM-GetS3Object
CPU type	cpu_type	SG-AWS-SSM-GetS3Object
Disk Space (GB)	disk_space	SG-AWS-SSM-GetS3Object SG-AWS-EC2
DNS Domain	dns_domain	SG-AWS-EC2
Fully qualified domain name	fqdn	SG-AWS-EC2
Host Name	host_name	SG-AWS-SSM-GetS3Object

Attribute label	Attribute name	Data source
Install Status	install_status	SG-AWS-EC2
Is Virtual	virtual	SG-AWS-EC2
Model ID	model_id	SG-AWS-SSM-GetS3Object
Object ID	object_id	SG-AWS-EC2
Operating System	os	SG-AWS-EC2 SG-AWS-Software-Inventory
Operational status	operational_status	SG-AWS-EC2
OS Version	os_version	SG-AWS-Software-Inventory
RAM (MB)	ram	SG-AWS-SSM-GetS3Object SG-AWS-EC2

Attribute label	Attribute name	Data source
Serial Number	serial_number	SG-AWS-SSM-GetS3Object

Note:

- If the AWS Systems Manager (SSM) service isn't enabled, the connector populates the server records in the Server [cmdb_ci_server] class. If the AWS SSM service is enabled, then based on the platform type obtained through the SSM service, the server records are populated in either the Linux Server [cmdb_ci_linux_server] class or the Windows Server [cmdb_ci_win_server] class. The Server [cmdb_ci_server] class is the parent class of the Linux Server [cmdb_ci_linux_server] and the Windows Server [cmdb_ci_win_server] classes.
- If the AWS SSM service is enabled, the name field is populated by the SG-AWS-Get-Inventory data source, and the sys_class_name field is populated by the SG-AWS-Software-Inventory data source. If the AWS SSM service isn't enabled, the name field is populated by the SG-AWS-EC2 data source, and the sys_class_name field isn't populated.

Relationships created for Server

Parent class	Relationship type	Child class
Server [cmdb_ci_server]	Virtualized by::Virtualizes	Virtual Machine Instance [cmdb_ci_vm_instance]
Server [cmdb_ci_server]	Owns::Owned by	IP Address [cmdb_ci_ip_address]
Server [cmdb_ci_server]	Owns::Owned by	Network Adapter [cmdb_ci_network_adapter]

Parent class	Relationship type	Child class
Server [cmdb_ci_server]	Reference	Software Installation [cmdb_sam_sw_install]
Server [cmdb_ci_server]	Reference	Key Value [cmdb_key_value]

Software [cmdb_ci_spkg]

The following attributes in the Software [cmdb_ci_spkg] table are populated by collected data when the Software Asset Management (SAM) application isn't installed:

Attribute label	Attribute name
Key	key
Discovery source	discovery_source
Name	name
Version	version
Manufacturer	manufacturer

Relationship created for Software

Parent class	Relationship type	Child class
Software [cmdb_ci_spkg]	Reference	Software Instance [cmdb_software_instance]

Software Installation [cmdb_sam_sw_install]

The following attributes in the Software Installation [cmdb_sam_sw_install] table are populated by collected data when the SAM application is installed:

Attribute label	Attribute name
Display name	display_name
Publisher	publisher
Version	version
Discovery source	discovery_source

Software Instance [cmdb_software_instance]

The following attributes in the Software Instance [cmdb_software_instance] table are populated by collected data when the SAM application isn't installed:

Attribute label	Attribute name
Name	name
Installed on	installed_on
Install date	install_date

Relationship created for Software Instance

Parent class	Relationship type	Child class
Software Instance [cmdb_software_instance]	Reference	Server [cmdb_ci_server]

Storage Mapping [cmdb_ci_storage_mapping]

The following attributes in the Storage Mapping [cmdb_ci_storage_mapping] table are populated by collected data:

Attribute label	Attribute name
Name	name

Attribute label	Attribute name
Object ID	object_id
Host	host
Install Status	install_status
Mapping Type	mapping_type
Mount Point	mount_point
Operational status	operational_status

Relationship created for Storage Mapping

Parent class	Relationship type	Child class
Storage Mapping [cmdb_ci_storage_ma pping]	Use End Point To::Use End Point From	Block Endpoint [cmdb_ci_endpoint_bl ock]

Storage Volume [cmdb_ci_storage_volume]

The following attributes in the Storage Volume [cmdb_ci_storage_volume] table are populated by collected data:

Attribute label	Attribute name
Object ID	object_id
Volume ID	volume_id
Install Status	install_status
Name	name
Operational status	operational_status
Size	size
Size bytes	size_bytes

Attribute label	Attribute name
State	state
Storage Type	storage_type

Relationships created for Storage Volume

Parent class	Relationship type	Child class
Storage Volume [cmdb_ci_storage_volume]	Hosted on::Hosts	AWS Datacenter [cmdb_ci_aws_datacenter]
Storage Volume [cmdb_ci_storage_volume]	Reference	Key Value [cmdb_key_value]

Storage Volume Snapshot [cmdb_ci_storage_vol_snapshot]

The following attributes in the Storage Volume Snapshot [cmdb_ci_storage_vol_snapshot] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Install Status	install_status
Operational status	operational_status

Relationship created for Storage Volume Snapshot

Parent class	Relationship type	Child class
Storage Volume Snapshot [cmdb_ci_storage_vol_snapshot]	Hosted on::Hosts	AWS Datacenter [cmdb_ci_aws_datacenter]

Virtual Machine Instance [cmdb_ci_vm_instance]

The following attributes in the Virtual Machine Instance [cmdb_ci_vm_instance] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
CPUs	cpus
Disks	disks
Disks size (GB)	disks_size
Fully qualified domain name	fqdn
Install Status	install_status
IP Address	ip_address
Memory (MB)	memory
Monitor	monitor
Network adapters	nics
Operational status	operational_status
Placement Group ID	placement_group_id
State	state
VM Instance ID	vm_inst_id

Relationships created for Virtual Machine Instance

Parent class	Relationship type	Child class
Virtual Machine Instance [cmdb_ci_vm_instance]	Located in::Houses	Availability Zone [cmdb_ci_availability_zone]
Virtual Machine Instance [cmdb_ci_vm_instance]	Hosted on::Hosts	AWS Datacenter [cmdb_ci_aws_datacenter]
Virtual Machine Instance [cmdb_ci_vm_instance]	Contains::Contained by	Storage Mapping [cmdb_ci_storage_mapping]
Virtual Machine Instance [cmdb_ci_vm_instance]	Provisioned From::Provisioned	Image [cmdb_ci_os_template]
Virtual Machine Instance [cmdb_ci_vm_instance]	Provisioned From::Provisioned	Cloud Hardware Type [cmdb_ci_cloud_hardware_type]
Virtual Machine Instance [cmdb_ci_vm_instance]	Provisioned From::Provisioned	Hardware Type [cmdb_ci_compute_template]
Virtual Machine Instance [cmdb_ci_vm_instance]	Reference	Key Value [cmdb_key_value]
Virtual Machine Instance	Use End Point To::Use End Point From	Storage Volume [cmdb_ci_storage_volume]

Parent class	Relationship type	Child class
[cmdb_ci_vm_instance]		
Virtual Machine Instance [cmdb_ci_vm_instance]	Use End Point To::Use End Point From	Cloud Mgmt Network Interface [cmdb_ci_nic]

VNIC Endpoint [cmdb_ci_endpoint_vnic]

The following attributes in the VNIC Endpoint [cmdb_ci_endpoint_vnic] table are populated by collected data:

Attribute label	Attribute name
Host	host
Name	name
Object ID	object_id
Install Status	install_status
Operational status	operational_status

Quick start tests for Service Graph Connector for AWS

Validate that Service Graph Connector for AWS still works after you make any configuration change such as applying an upgrade or developing an application. Copy and customize these quick start tests to pass when using your instance-specific data.

Danger: By default, the system property that is used to run automated tests is disabled to prevent you from accidentally running these tests on a production system. To avoid data corruption or an outage, run tests only on development, test, and other non-production instances. See [Enable or disable executing Automated Test Framework tests](#).

SG-AWS: Service Graph Connector for AWS Validation test suite

Test suite to verify the integrity of an integration using multiple tests.

Test	Description	Release version
SG-AWS API Gateway	Validate the ingestion of multiple CI records in the Cloud Gateway.	Utah
SG-AWS DynamoDb	Validate the ingestion of multiple CI records in the DynamoDB table.	Utah
SG-AWS EC2	Loads data in the CMDB such as all dependent CIs such as Organizations, Service-Accounts, Networks, Subnets, and Network-Interfaces, after which mock data for SG-AWS-EC2 is prepared.	Utah
SG-AWS EKS Cluster	Validates the ingestion of multiple CI records in the Kubernetes Cluster table.	Utah
SG-AWS ELB V1	Validates the ingestion of multiple CI records in the Cloud Load Balancer.	Utah
SG-AWS ELB V2	Validates the ingestion of multiple CI records in the Cloud Load Balancer.	Utah

Test	Description	Release version
SG-AWS Generic Resource	Validates the ingestion of multiple CI records in the Cloud Resource table.	Utah
SG-AWS Hardware Type	Validates the ingestion of multiple CI records in the Hardware Type table.	Utah
SG-AWS Image Id	Validates the ingestion of multiple CI records in the Image table.	Utah
SG-AWS Image Private	Validates the ingestion of multiple CI records in the Image table.	Utah
SG-AWS Lambda	Validates the ingestion of multiple CI records in the Cloud function table.	Utah
SG-AWS Network Interface	Validates the ingestion of multiple CI records in the Cloud Mgmt Network Interface table.	Utah
SG-AWS Organization	Tests RTE mappings for the ingestion of CIs in the Cloud Organization table.	Utah
SG-AWS RDS	Validates the ingestion of multiple CI records in the Cloud Database table.	Utah
SG-AWS S3	Validates the ingestion of multiple CI records	Utah

Test	Description	Release version
	in the Cloud Object Storage table.	
SG-AWS Security Group	Validates the ingestion of multiple CI records in the Compute Security Group table.	Utah
SG-AWS Service Account	Tests RTE mappings for the ingestion of CIs in the Cloud Service Account table.	Utah
SG-AWS Software Inventory	Validates the ingestion of Software Data and classification of CI records in the Server table. It tests whether the server is classified as Linux or Windows.	Utah
SG-AWS Software Remove	Validates the removal of Software Data from the cmdb_software_instance or the cmdb_sam_sw_install table.	Utah
SG-AWS Storage Volume	Loads mock import data for Storage Volume into the relevant staging table. Validates if fields are populated as expected in the cmdb_ci_storage_volume, cmdb_ci_storage_vol_snapshot, and sn_aws_extension_attri	Utah

Test	Description	Release version
	butes tables and if the appropriate relationships have been created between the inserted CIs.	
SG-AWS SSM SendCommand	Validates the ingestion of multiple CI records in the TCP Connection and Running Process tables. Updates records in the Server table.	Utah
SG-AWS Subnets	Validates the ingestion of multiple CIs in the Cloud Subnet table.	Utah
SG-AWS VM Hw Consolidation	Validates the update of multiple CI records in the Virtual Machine Instance and Server table.	Utah
SG-AWS VPC	Tests RTE mappings associated with SG-AWS VPC Data source. It loads import set data for VPCs, Datacenters, Service-Account, and Organizations. Validates if the CIs have been added to the cmdb_ci_network table, the fields are populated with the expected value, and appropriate relationships verified	Utah

Test	Description	Release version
	through the cmdb_rel_ci table.	

Related topics

- [Quick start tests](#)

Accessing the connection details of Service Graph Connector for AWS

You can access the connection details of the Service Graph Connector for AWS in a single view using the common connection framework (CCF) included within the Integration Commons for CMDB (sn_cmdb_int_util) store app.

With the CCF, you can access all the connections used by the Service Graph Connector for AWS. The connection details include the connection alias, connection properties, data sources, and scheduled data imports associated with a connection. You can also test the connection. For more information, see [Accessing the connection details of Service Graph Connectors](#).

Access the details of an AWS connection

Access the details of an AWS connection configured for the Service Graph Connector for AWS.

Before you begin

Role required: admin

Procedure

1. Navigate to **All > Service Graph Connectors > AWS > Connections**.
2. From the **Name** column of the Service Graph Connections list, select a connection.
3. On the Service Graph Connections page, view the connection details such as the connection name, alias, and data sources.

4. Select a related list to view further details of the connection.
5. (Optional) Select the **Test Connection** related link to test the connection.

Additional features within the Service Graph Connector for AWS

Use the additional features available within the Service Graph Connector for AWS to maximize its use for importing data.

The following features are available within the Service Graph Connector for AWS:

Enabling optimized deletion of retired CIs in AWS

Enable the `sn_aws_integ.enableOptimizedDeletion` system property for the optimized deletion of retired configuration items (CIs) in AWS during delta import of data.

Enabling optimized SG-AWS-SendCommand data source performance

When using the SG-AWS-Software-Inventory data source, enable the `sn_aws_integ.is_software_inventory_enabled` system property for optimizing the performance of the SG-AWS-SendCommand and SG-AWS-SSM-GetS3Object data sources.

Handling import failures for large AWS resources

Manage the data source failure when importing resources from AWS.

Handling memory utilization errors for Service Graph Connector for AWS

Avoid the memory utilization errors when using the Service Graph Connector for AWS to import a large number of CIs from an AWS instance.

Setting up partition size with parallel loading enabled

Optimize the performance of the SG-AWS-Image-Private, SG-AWS-Tags, SG-AWS-SendCommand, and AWS-SSM-GetS3Object data sources by configuring the partition size setting in the `aws_account_partition_size` connection property.

Disable parallel loading for Service Graph Connector for AWS data sources

Disable parallel loading for specific data sources in the Service Graph Connector for AWS.

Handling parallel deletion of CIs in Service Graph Connector for AWS

Optimize performance when using the Service Graph Connector for AWS by enabling the parallel deletion of CIs.

Enable mapping for Cloud Os Image in AWS

Enable mapping of AWS Cloud Os Image data to the Cloud Image [cmdb_ci_cloud_os_image] table.

Enable optimized deletion of retired CIs in AWS

Enable the `sn_aws_integ.enableOptimizedDeletion` system property for the optimized deletion of retired configuration items (CIs) in AWS during full import of data.

Before you begin

Set the application scope to Service Graph Connector for AWS using the application picker. For more information, see [Application picker](#).

Role required: admin

About this task

By default, the `ListDiscoveredResources` API is used in batches of 20 to ascertain deleted resource data as there's no other API in AWS that provides deleted resource deltas. This action adversely impacts performance, particularly with a large number of resources, as the majority of time is dedicated to API calls.

When your AWS instance has a large number of CIs to import, you can enable the `sn_aws_integ.enableOptimizedDeletion` property to accelerate the import time. The property enables deleting any retired CI classes with an improved logic. In this optimized deletion approach, a complete pull is conducted. Without depending on AWS API calls, the deleted CI is determined by comparing imported resources from the last

run with the resources imported in the current run. By default, the value of the property is set to `false`.

Procedure

1. Enter `sys_properties.list` in the navigation filter, and then open the `sn_aws_integ.enableOptimizedDeletion` property.
2. In the **Value** field, enter `true`.
3. Select **Update**.

Enable optimized SG-AWS-SendCommand data source performance

When the SG-AWS-Software-Inventory data source is being used, enable the `sn_aws_integ.is_software_inventory_enabled` system property for optimizing the performance of the SG-AWS-SendCommand and SG-AWS-SSM-GetS3Object data sources.

Before you begin

Set the application scope to Service Graph Connector for AWS using the application picker. For more information, see [Application picker](#).

Role required: admin

About this task

The SG-AWS-SendCommand and SG-AWS-SSM-GetS3Object data sources utilize SSM GetInventory API calls for importing software data, impacting performance. To enhance the performance of the SG-AWS-SendCommand and SG-AWS-SSM-GetS3Object data sources, consider leveraging the SG-AWS-Software-Inventory data source by enabling the `sn_aws_integ.is_software_inventory_enabled` system property. This optimization skips a substantial number of SSM GetInventory API calls, enhancing the performance of the SG-AWS-SendCommand and SG-AWS-SSM-GetS3Object data sources. However, the SG-AWS-SendCommand and SG-AWS-SSM-GetS3Object data sources import data from the same servers that are configured for the SG-AWS-Software-Inventory data source.

By default, the `sn_aws_integ.is_software_inventory_enabled` property is set to `false`. To enhance the performance of the SG-AWS-SendCommand and SG-AWS-SSM-GetS3Object data sources, set the `sn_aws_integ.is_software_inventory_enabled` property to `true`.

Procedure

1. Enter `sys_properties.list` in the navigation filter, and then open the `sn_aws_integ.is_software_inventory_enabled` property.
2. In the **Value** field, enter `true`.
3. Select **Update**.

Handling import failures for large AWS resources

Manage the data source failure when importing resources from AWS.

When importing large AWS resources, the data sources in the Service Graph Connector for AWS might fail because of a Direct REST response exceeding the 1024-MB limit. For more information, see [REST step](#).

You can save the response as an attachment to handle the REST response size limitation error in the Service Graph Connector for AWS.

Save the response as an attachment

With the save response as an attachment feature, the response of each data source is temporarily stored as an attachment and after the response is read, the attachment is deleted automatically. The save response as an attachment feature is activated by default and is triggered by the `sn_aws_integ.save_payload_as_attachment` system property.

As a user with the admin role, you can deactivate the save response as an attachment feature by setting the `sn_aws_integ.save_payload_as_attachment` property to `false`. When the feature is deactivated and the response size is within the limit, the data sources run normally.

Handling memory utilization errors for Service Graph Connector for AWS

You can use the `SgAwsApplicationProperties` script include to avoid memory utilization errors when importing a large number of configuration items (CIs) from an AWS instance via the Service Graph Connector for AWS.

By making multiple database calls to ingest AWS data in smaller batches, the `SgAwsApplicationProperties` script include avoids overloading of the Robust Transform Engine (RTE) and results in reduced memory usage. However, for the script include to function properly, verify that the AWS Config aggregator is enabled on your AWS instance.

To access the script include, navigate to **All > System Definitions > Script Includes**, and then search for and select `SgAwsApplicationProperties`. For more information, see [Script includes](#).

In the `SgAwsApplicationProperties` script include, you can modify the functions as described in the following table to resolve memory utilization errors.

`SgAwsApplicationProperties` script include functions

Functionality	Implementation
Enable multiple database calls to import the AWS Config data.	Set the value of the <code>isConfigDbLoadEnabled</code> function to <code>true</code>. For the function to work correctly, verify that the AWS Config aggregator is enabled. Note: Enabling <code>isConfigDbLoadEnabled</code> might impact the import set load time as multiple database calls are being made instead of a single database call. Enabling the <code>isConfigDbLoadEnabled</code> function resolves the memory utilization

Functionality	Implementation
	error for the following data sources: • SG-AWS-API-Gateway • SG-AWS-DynamoDb • SG-AWS-EC2 • SG-AWS-EKS-Cluster • SG-AWS-ELB-V1 • SG-AWS-ELB-V2 • SG-AWS-Generic-Resources • SG-AWS-Lambda • SG-AWS-Network-Interface • SG-AWS-RDS • SG-AWS-S3 • SG-AWS-Security-Group • SG-AWS-Storage-Volume • SG-AWS-Subnets • SG-AWS-VPC
Enable multiple database calls to import the tag-related data from your AWS service account.	Set the value of the <code>isConfigTagsDbLoadEnabled</code> function to <code>true</code>. For the function to work correctly, verify that both the AWS Config aggregator and the <code>isConfigDbLoadEnabled</code> function are enabled. Enabling the <code>isConfigTagsDbLoadEnabled</code> function facilitates the AWS Config

Functionality	Implementation
	aggregator to reduce the load time for tag-related data in the following data sources: • SG-AWS-Lambda • SG-AWS-DynamoDb • SG-AWS-ELB-V1 • SG-AWS-ELB-V2

Set partition size for parallel loading-enabled AWS accounts

When using the SG-AWS-Image-Private, SG-AWS-Tags, SG-AWS-SendCommand, and AWS-SSM-GetS3Object data sources, you can optimize their performance by setting the partition size in the `aws_account_partition_size` connection property.

Before you begin

Set the application scope to Service Graph Connector for AWS using the application picker. For more information, see [Application picker](#).

Role required: admin

About this task

The parallel loading feature is available for the SG-AWS-Image-Private, SG-AWS-Tags, SG-AWS-SendCommand, and AWS-SSM-GetS3Object data sources from the Washington DC Patch 5 release. This feature determines the number of AWS accounts that can be executed as parallel jobs to retrieve configuration item (CI) data. Partition size determines how many AWS accounts a single parallel-thread evaluates for data retrieval. For example, if you set the `aws_account_partition_size` property to 5 and there are 25 AWS accounts, the application generates 5 parallel jobs during data loading.

Note: If the `aws_account_partition_size` property isn't configured, the partition size computation feature of the Integration Commons for CMDb application automatically determines the partition size for parallel loading at runtime. See [Partition size computation for parallel loading in Integration Commons for CMDb](#).

Procedure

1. Navigate to **All > Service Graph Connectors > AWS > Connections**.
2. From the **Name** column of the Service Graph Connections list, select a connection.
3. From the **Property** column of the Service Graph Connection Properties related list, select **aws_account_partition_size**.
4. In the **Value** field, enter a numeric value greater than 0 for the partition size.
5. Select **Update**.

Disable parallel loading for Service Graph Connector for AWS data sources

You can disable parallel loading for specific data sources in the Service Graph Connector for AWS.

Before you begin

Set the application scope to Service Graph Connector for AWS using the application picker. For more information, see [Application picker](#).

Role required: admin

Procedure

1. Navigate to **All > Service Graph Connectors > AWS > Data sources**.
2. From the list of data sources, open the data source for which you want to disable parallel loading.
3. Clear the **Enable parallel loading** check box.

4. Select **Update**.

Handling parallel deletion of CIs in Service Graph Connector for AWS

Parallel deletion of configuration items (CIs) in the Service Graph Connector for AWS optimizes performance.

The parallel deletion feature is available for the following data sources from Washington DC patch 5 after the parallel job plugin is activated:

- SG-AWS-API-Gateway
- SG-AWS-DynamoDb
- SG-AWS-EC2
- SG-AWS-ELB-V1
- SG-AWS-ELB-V2
- SG-AWS-Generic-Resources
- SG-AWS-Lambda
- SG-AWS-Network-Interface
- SG-AWS-RDS
- SG-AWS-S3
- SG-AWS-Security-Group
- SG-AWS-Storage-Volume
- SG-AWS-Subnets
- SG-AWS-VPC

Note: If you upgraded to Service Graph Connector for AWS version 2.9.1 before the parallel job plugin was activated, you must run the SGC - Parallel Loading Fix script to enable parallel deletion.

Multiple jobs are created for deleting CIs based on the number of accounts in your configuration. You can modify the number of jobs that

are created by setting the partition size in the `aws_account_partition_size` connection property. See [Set partition size for parallel loading-enabled AWS accounts](#).

Note: Only one job is created to import new and updated CIs.

Enable mapping for Cloud Os Image in AWS

Enable the `sn_cmdb_ci_class.use_single_cloud_os_image` system property to add Cloud Os Image data to the Cloud Image `[cmdb_ci_cloud_os_image]` table instead of adding the data to the Image `[cmdb_ci_os_template]` table.

Before you begin

Set the application scope to Service Graph Connector for AWS using the application picker. For more information, see [Application picker](#).

Role required: admin

About this task

In Service Graph Connector for AWS version 2.9.1 and earlier, the Cloud Os Image data was added to the Image `[cmdb_ci_os_template]` table. Multiple records were created for the same image because the Image `[cmdb_ci_os_template]` table is a dependent class. Starting with the Service Graph Connector for AWS 2.10.0 version, Cloud Os Image data is added to the Cloud Image `[cmdb_ci_cloud_os_image]` table, which is an independent class, when the `sn_cmdb_ci_class.use_single_cloud_os_image` system property is enabled.

Procedure

1. Enter `sys_properties.list` in the navigation filter, and then open the `sn_cmdb_ci_class.use_single_cloud_os_image` property.
2. In the **Value** field, enter `true`.
3. Select **Update**.

Service Graph Connector for AWS reference

Reference topics provide additional information about Service Graph Connector for AWS components.

- [Supported AWS resource types](#)

Several AWS resource types are imported as CMDB data by the Service Graph Connector for AWS.

- [AWS discovery solutions comparison](#)

ITOM Visibility applications discover a variety of AWS resources and populate the relevant configuration item (CI) classes in the Configuration Management Database (CMDB) with their attributes.

- [Life cycle management of records in Service Graph Connector for AWS](#)

Life cycle management in the Service Graph Connector for AWS monitors and updates the statuses of AWS resources throughout their entire life cycle, from creation to deletion.

- [Record removal process in Service Graph Connector for AWS](#)

The Service Graph Connector for AWS uses the optimized deletion approach to manage removal of records.

Supported AWS resource types

Several AWS resource types are imported as CMDB data by the Service Graph Connector for AWS.

AWS Config resources

When the AWS Config aggregator is enabled, the following APIs are invoked:

- [ListDiscoveredResources](#)
- [BatchGetAggregateResourceConfig](#)
- [SelectAggregateResourceConfig](#)

When the AWS Config aggregator is disabled, the following APIs are invoked:

- [ListDiscoveredResources](#)
- [SelectResourceConfig](#)
- [BatchGetResourceConfigAction](#)

Note: For a list of AWS resource types supported by AWS Config, see [Supported Resource Types](#) on the AWS documentation site.

CMDB CI classes and AWS Config resources

CMDB CI classes	AWS resource type	AWS Config resource identifier
Cloud Network [cmdb_ci_network]	Amazon VPC	AWS::EC2::VPC
Cloud Subnet [cmdb_ci_cloud_subnet]	Amazon subnet	AWS::EC2::Subnet
Cloud Mgmt Network Interface [cmdb_ci_nic]	Amazon Elastic network interface	AWS::EC2::NetworkInterface
Compute Security Group [cmdb_ci_compute_security_group]	Amazon EC2 security group	AWS::EC2::SecurityGroup
Storage Volume [cmdb_ci_storage_volume]	Amazon EBS volume	AWS::EC2::Volume

CMDB CI classes	AWS resource type	AWS Config resource identifier
Cloud Load Balancer [cmdb_ci_cloud_load_balancer]	Classic load balancer	AWS::ElasticLoadBalancing::LoadBalancer
Cloud Load Balancer [cmdb_ci_cloud_load_balancer]	Application load balancer, network load balancer, or gateway load balancer	AWS::ElasticLoadBalancingV2::LoadBalancer
Cloud Gateway [cmdb_ci_cloud_gateway]	API gateway with WebSocket APIs or HTTP APIs	AWS::ApiGatewayV2::Api
Virtual Machine Instance [cmdb_ci_vm_instance]	Amazon EC2 instance	AWS::EC2::Instance
Server [cmdb_ci_server]	Amazon EC2 instance	AWS::EC2::Instance
Cloud DataBase [cmdb_ci_cloud_database]	Amazon RDS DB instance	AWS::RDS::DBInstance
Cloud Function [cmdb_ci_cloud_function]	Lambda function	AWS::Lambda::Function

CMDB CI classes	AWS resource type	AWS Config resource identifier
Cloud Object Storage [cmdb_ci_cloud_object_storage]	Amazon S3 bucket	AWS::S3::Bucket
DynamoDB Table [cmdb_ci_dynamodb_table]	DynamoDB table	AWS::DynamoDB::Table
Software [cmdb_ci_spkg]	Software configuration for managed instances	AWS::SSM::ManagedInstanceInventory
Kubernetes Cluster [cmdb_ci_kubernetes_cluster]	Amazon EKS cluster	AWS::EKS::Cluster
Amazon Redshift [cmdb_ci_aws_redshift]	Amazon Redshift	AWS::Redshift::Cluster

Amazon EC2 resources

CMDB CI classes and Amazon EC2 resources

CMDB CI classes	AWS resource type	AWS API action
Availability Zone [cmdb_ci_availability_zone]	AWS availability zone	None

CMDB CI classes	AWS resource type	AWS API action
Hardware Type [cmdb_ci_compute_template]	Amazon EC2 instance types	DescribeInstanceTypes
Image [cmdb_ci_os_template]	Amazon Machine Image (AMI)	DescribeImages
Cloud Image [cmdb_ci_cloud_os_image]	Amazon Machine Image (AMI)	DescribeImages
Cloud Hardware Type [cmdb_ci_cloud_hardware_type]	Amazon EC2 instance types	DescribeInstanceTypes

AWS Systems Manager resources

All AWS Systems Manager resources invoke the [SendCommand](#) API action.

CMDB CI classes and AWS Systems Manager resources

CMDB CI classes	AWS resource type	AWS API action
Running Process [cmdb_running_processes]	Running process in EC2 instances	SendCommand

CMDB CI classes	AWS resource type	AWS API action
TCP Connections [cmdb_tcp]	TCP connections in EC2 instances	
Kubernetes Node [cmdb_ci_kubernetes_node]	Amazon EKS node	
Kubernetes Pod [cmdb_ci_kubernetes_pod]	Amazon EKS pod	
Kubernetes Service [cmdb_ci_kubernetes_service]	Amazon EKS service	
Kubernetes DaemonSet [cmdb_ci_kubernetes_daemonset]	Kubernetes DaemonSet	
Kubernetes Namespace [cmdb_ci_kubernetes_namespace]	Kubernetes namespace	
Kubernetes Deployment	Kubernetes deployment	

CMDB CI classes	AWS resource type	AWS API action
[cmdb_ci_kubernetes_deployment]		
Docker Container [cmdb_ci_docker_container]	Docker container	
Kubernetes Volume [cmdb_ci_kubernetes_volume]	Kubernetes Volume	

Organizations resources

CMDB CI classes and organizations

CMDB CI classes	AWS resource type	AWS API action
Cloud Organizations [cmdb_ci_cloud_org]	AWS Organizations	DescribeOrganization
Cloud Service Account [cmdb_ci_cloud_service_account]	AWS accounts	ListAccounts
AWS Datacenter [cmdb_ci_aws_datacenter]	AWS regions	DescribeRegions

Resource Group Tagging resources

CMDB CI class and resource group tagging

CMDB classes	AWS resource type	AWS API action
Key Value [cmdb_key_value]	Tags for AWS Lambda, Amazon DynamoDB, ELBv1, and ELBv2	GetResources

Generic resources

Generic resources are stored in the Cloud Resource [cmdb_ci_cmp_resource] CMDB CI class.

CMDB CI classes and generic resources

CMDB CI class	AWS resource type	AWS API action
Cloud Resource [cmdb_ci_cmp_resource]	State Manager association compliance	AWS::SSM::Association Compliance
	Database subnet group	AWS::RDS::DBSubnetGroup
	AWS WAF rate based rule	AWS::WAF::RateBased Rule
	AWS WAF regional rule group	AWS::WAFRegional::RuleGroup

CMDB CI class	AWS resource type	AWS API action
	AWS WAF regional rate based rule	AWS::WAFRegional::RateBasedRule
	Managed policies and inline policies	AWS::IAM::Policy
	AWS WAF Web ACL rule	AWS::WAF::WebACL
	AWS WAF Regional Web ACL rule	AWS::WAFRegional::WebACL
	AWS Certificate Manager (ACM) certificate	AWS::ACM::Certificate
	Amazon EC2 Auto Scaling group	AWS::AutoScaling::AutoScalingGroup
	IAM user for AWS account	AWS::IAM::User
	Amazon Redshift event subscription	AWS::Redshift::EventSubscription

CMDB CI class	AWS resource type	AWS API action
	Amazon RDS DB snapshot	AWS::RDS::DBSnapshot
	Amazon Redshift cluster parameter group	AWS::Redshift::ClusterParameterGroup
	Amazon EC2 Auto Scaling group launch configuration	AWS::AutoScaling::LaunchConfiguration
	Amazon Redshift cluster snapshot	AWS::Redshift::ClusterSnapshot
	AWS CloudFormation stack	AWS::CloudFormation::Stack
	AWS CodeBuild project	AWS::CodeBuild::Project
	AWS Elastic Beanstalk application version	AWS::ElasticBeanstalk::ApplicationVersion
	AWS CodePipeline pipeline	AWS::CodePipeline::Pipeline

CMDB CI class	AWS resource type	AWS API action
	Amazon Redshift security group	AWS::Redshift::ClusterSecurityGroup
	Amazon CloudFront distribution	AWS::CloudFront::Distribution
	AWS CloudTrail trail	AWS::CloudTrail::Trail
	Amazon CloudFront streaming distribution	AWS::CloudFront::StreamingDistribution
	AWS WAF rule group	AWS::WAF::RuleGroup
	Amazon RDS DB security group	AWS::RDS::DBSecurityGroup
	Amazon RDS event subscription	AWS::RDS::EventSubscription
	AWS WAF regional rule	AWS::WAFRegional::Rule
	AWS WAF rule	AWS::WAF::Rule

CMDB CI class	AWS resource type	AWS API action
	IAM role for AWS account	AWS::IAM::Role
	Amazon EC2 Auto Scaling scheduled action	AWS::AutoScaling::ScheduledAction
	AWS Elastic Beanstalk environment	AWS::ElasticBeanstalk::Environment
	AWS Redshift cluster	AWS::Redshift::Cluster
	Amazon EC2 Auto Scaling group scaling policy	AWS::AutoScaling::ScalingPolicy
	AWS Shield protected AWS resource	AWS::Shield::Protection
	Amazon CloudWatch alarm	AWS::CloudWatch::Alarm
	EncryptionConfig - AWS X-Ray	AWS::XRay::EncryptionConfig

CMDB CI class	AWS resource type	AWS API action
	AWS Shield protected AWS region	AWS::ShieldRegional::Protection
	Amazon Redshift subnet group	AWS::Redshift::ClusterSubnetGroup
	AWS Elastic Beanstalk application	AWS::ElasticBeanstalk::Application
	IAM group for AWS account	AWS::IAM::Group

AWS discovery solutions comparison

ITOM Visibility applications discover a variety of AWS resources and populate the relevant configuration item (CI) classes in the Configuration Management Database (CMDB) with their attributes.

ITOM Visibility solutions

ITOM Visibility can offer you comprehensive visibility into your organization's cloud infrastructure. Each application offers unique strengths and capabilities, and by comparing them you can gain holistic insights on their capabilities and benefits.

ITOM Visibility solutions for discovering AWS

ITOM Visibility solution	Overview	Configuration
Discovery and Service Mapping Patterns and Visibility Content	Discovery patterns used by ITOM Visibility	You must update your store apps to

ITOM Visibility solution	Overview	Configuration
		the latest release version. You must provide ITOM Visibility with access permissions. For more information, see Setting up AWS service accounts
Service Graph Connector for AWS	Service Graph Connector for AWS	Configure Service Graph Connector for AWS using the guided setup

Note:

The CI class, in most cases, is applicable to both solutions. If it's not applicable to one of the solutions, the solution is marked "N/A".

Discovery and Service Mapping Patterns vs Service Graph Connector for AWS

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
AWS::AppSync::API	Yes	No	cmdb_aws_appsync_api
AWS::Athena::Workgroup	Yes	No	cmdb_aws_athena_workgroup
AWS::Backup::BackupPlan	Yes	No	cmdb_aws_backup_backup_plan

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
AWS::Backup::BackupVault	Yes	No	cmdb_aws_backup_backup_vault
AWS::CloudFront::Distribution	Yes	N/A	cmdb_aws_cloudfront_distribution
AWS::CloudFront::Distribution	N/A	Yes	cmdb_ci_cmp_resource
AWS::CloudWatch::Log	Yes	No	cmdb_aws_cloudwatch_log
AWS::CodeDeployment::Deployment	Yes	No	cmdb_aws_codedeploy_deployment
AWS::CodePipeline::Pipeline	Yes	N/A	cmdb_aws_codepipeline_pipeline
AWS::CodePipeline::Pipeline	N/A	Yes	cmdb_ci_cmp_resource
AWS::DMS::Endpoint	Yes	No	cmdb_aws_dms_endpoint
AWS::ElasticBeanstalk::Application	Yes	N/A	cmdb_aws_elastic_beanstalk_application
AWS::ElasticBeanstalk::Application	N/A	Yes	cmdb_ci_cmp_resource

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
AWS::EMR::Cluster	Yes	No	cmdb_aws_emr_cluster
AWS::EventBridge::EventBus	Yes	No	cmdb_aws_eventbridge_event_bus
AWS::FSx::Backup	Yes	No	cmdb_aws_fsx_backup
AWS::FSx::FileSystem	Yes	No	cmdb_aws_fsx_file_system
AWS::Glue::Database	Yes	No	cmdb_aws_glue_database
AWS::IAM::Policy	Yes	N/A	cmdb_aws_iam_policy
AWS::IAM::Policy	N/A	Yes	cmdb_ci_cmp_resource
AWS::IAM::Role	Yes	N/A	cmdb_aws_iam_role
AWS::IAM::Role	N/A	Yes	cmdb_ci_cmp_resource
AWS::IAM::User	Yes	N/A	cmdb_aws_iam_user
AWS::IAM::User	N/A	Yes	cmdb_ci_cmp_resource
AWS::KMS::Key	Yes	No	cmdb_aws_kms_key

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
AWS::MQ::Broker	Yes	No	cmdb_aws_mq_broker
AWS::MQ::Configuration	Yes	No	cmdb_aws_mq_configuration
AWS::S3::Glacier Vault	Yes	No	cmdb_aws_s3_glacier_vault
AWS::SageMaker::NotebookInstance	Yes	No	cmdb_aws_sagemaker_notebook_instance
AWS::SecretsManager::Secret	Yes	No	cmdb_aws_secrets_manager_secret
AWS::SimpleEmailService::Identity	Yes	No	cmdb_aws_simple_email_service_identity
AWS::SQS::Queue	Yes	No	cmdb_aws_sqs_queue
AWS::StepFunctions::StateMachine	Yes	No	cmdb_aws_step_functions_state_machine
AWS::StorageGateway::Gateway	Yes	No	cmdb_aws_storage_gateway_gateway
AWS::SystemsManager::Document	Yes	No	cmdb_aws_systems_manager_document

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
AWS::SystemsManager::ParameterStore	Yes	No	cmdb_aws_systems_manager_parameter_store
AWS::TransferFamily::Server	Yes	No	cmdb_aws_transfer_family_server
AWS::VPC::FlowLog	Yes	No	cmdb_aws_vpc_flow_log
AWS::VPC::ManagedPrefixList	Yes	No	cmdb_aws_vpc_managed_prefix_list
AWS::XRay::SamplingRule	Yes	No	cmdb_aws_xray_sampling_rule
ec2::DescribeAvailabilityZones	Yes	Yes	cmdb_ci_availability_zone
ec2::DescribeRegions	Yes	Yes	cmdb_ci_aws_datacenter
AWS::Kinesis::Stream	Yes	No	cmdb_ci_aws_kinesis_data
AWS::KinesisAnalytics	Yes	No	cmdb_ci_aws_kinesis_data_analytics
AWS::KinesisFirehose	Yes	No	cmdb_ci_aws_kinesis_firehose
AWS::KinesisVideo	Yes	No	cmdb_ci_aws_kinesis_video

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
AWS::Redshift::Cluster	Yes	N/A	cmdb_ci_aws_redshift
AWS::Redshift::Cluster	N/A	Yes	cmdb_ci_cmp_resource
AWS::CertificateManager::Certificate	Yes	No	cmdb_ci_certificate
AWS::Cognito::UserPool	Yes	No	cmdb_ci_cloud_authentication
AWS::RDS::DBInstance	Yes	Yes	cmdb_ci_cloud_database
AWS::ECS::Cluster	Yes	No	cmdb_ci_cloud_ecs_cluster
AWS::ECS::Service	Yes	No	cmdb_ci_cloud_ecs_service
AWS::ECS::TaskDefinition	Yes	No	cmdb_ci_cloud_ecs_task_definition
AWS::Lambda::Function	Yes	Yes	cmdb_ci_cloud_function
AWS::ApiGateway::RestApi	Yes	Yes	cmdb_ci_cloud_gateway
ec2::DescribeInstanceTypes	Yes	Yes	cmdb_ci_cloud_hardware_type
AWS::EC2::Host	Yes	No	cmdb_ci_cloud_host

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
AWS::EC2::KeyPair	Yes	No	cmdb_ci_cloud_key_pair
AWS::ElasticLoadBalancingV2::LoadBalancer	Yes	Yes	cmdb_ci_cloud_load_balancer
Amazon::S3::Bucket	Yes	Yes	cmdb_ci_cloud_object_storage
Organizations	Yes	Yes	cmdb_ci_cloud_org or cmdb_ci_cloud_service_account
Organizations::AWS accounts	Yes	Yes	cmdb_ci_cloud_service_account
AWS::EC2::Subnet	Yes	Yes	cmdb_ci_cloud_subnet
AWS SSM agents	Yes	No	cmdb_ci_cloud_system_management_agent
AWS::EC2::Subnet	Yes	No	cmdb_ci_cloud_subnet
AWS::WAF::WebACL	N/A	Yes	cmdb_ci_cmp_resource
AWS::WAFRegional::WebACL	N/A	Yes	cmdb_ci_cmp_resource
AWS::ACM::Certificate	N/A	Yes	cmdb_ci_cmp_resource

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
AWS::Redshift::EventSubscription	N/A	Yes	cmdb_ci_cmp_resource
AWS::RDS::DBSnapshot	N/A	Yes	cmdb_ci_cmp_resource
AWS::Redshift::ClusterParameterGroup	N/A	Yes	cmdb_ci_cmp_resource
AWS::AutoScaling::LaunchConfiguration	N/A	Yes	cmdb_ci_cmp_resource
AWS::Redshift::ClusterSnapshot	N/A	Yes	cmdb_ci_cmp_resource
AWS::CloudFormation::Stack	N/A	Yes	cmdb_ci_cmp_resource
AWS::CodeBuild::Project	N/A	Yes	cmdb_ci_cmp_resource
AWS::ElasticBeanstalk::ApplicationVersion	N/A	Yes	cmdb_ci_cmp_resource
AWS::Redshift::ClusterSecurityGroup	N/A	Yes	cmdb_ci_cmp_resource
AWS::CloudTrail::Trail	N/A	Yes	cmdb_ci_cmp_resource
AWS::CloudFront::StreamingDistribution	N/A	Yes	cmdb_ci_cmp_resource

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
AWS::WAF::Rule Group	N/A	Yes	cmdb_ci_cmp_resource
AWS::RDS::DBSecurityGroup	N/A	Yes	cmdb_ci_cmp_resource
AWS::RDS::Event Subscription	N/A	Yes	cmdb_ci_cmp_resource
AWS::WAF::Rule	N/A	Yes	cmdb_ci_cmp_resource
AWS::AutoScaling::ScheduledAction	N/A	Yes	cmdb_ci_cmp_resource
AWS::ElasticBeanstalk::Environment	N/A	Yes	cmdb_ci_cmp_resource
AWS::AutoScaling::ScalingPolicy	N/A	Yes	cmdb_ci_cmp_resource
AWS::Shield::Protection	N/A	Yes	cmdb_ci_cmp_resource
AWS::CloudWatch::Alarm	N/A	Yes	cmdb_ci_cmp_resource
AWS::XRay::EncryptionConfig	N/A	Yes	cmdb_ci_cmp_resource
AWS::ShieldRegional::Protection	N/A	Yes	cmdb_ci_cmp_resource

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
AWS::Redshift::ClusterSubnetGroup	N/A	Yes	cmdb_ci_cmp_resource
AWS::IAM::Group	N/A	Yes	cmdb_ci_cmp_resource
AWS::WAFRegional::RateBasedRule	N/A	Yes	cmdb_ci_cmp_resource
AWS::WAFRegional::RuleGroup	N/A	Yes	cmdb_ci_cmp_resource
AWS::WAF::RateBasedRule	N/A	Yes	cmdb_ci_cmp_resource
AWS::RDS::DBSubnetGroup	N/A	Yes	cmdb_ci_cmp_resource
AWS::SSM::AssociationCompliance	N/A	Yes	cmdb_ci_cmp_resource
AWS::EC2::SecurityGroup	Yes	Yes	cmdb_ci_compute_security_group
ec2:DescribeInstanceTypes	Yes	Yes	cmdb_ci_compute_template
AWS::ECR::Repository	Yes	No	cmdb_ci_container_repository
AWS::EC2::CustomerGateway	Yes	No	cmdb_ci_customer_gateway

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
Docker container	Yes	Yes	cmdb_ci_docker_container
ecr:DescribeImages	Yes	No	cmdb_ci_docker_image
AWS::DynamoDB::GlobalTable	Yes	Yes	cmdb_ci_dynamodb_table
AWS::AutoScaling::AutoScalingGroup	Yes	N/A	cmdb_ci_instance_scale_set
AWS::AutoScaling::AutoScalingGroup	N/A	Yes	cmdb_ci_compute_resource
AWS::EC2::InternetGateway	Yes	No	cmdb_ci_internet_gateway
AWS::EKS::Cluster	Yes	Yes	cmdb_ci_kubernetes_cluster
AWS::ElasticLoadBalancingV2::TargetGroup	Yes	No	cmdb_ci_lb_pool
AWS::ElasticLoadBalancingV2::TargetGroupTargetDescription	Yes	No	cmdb_ci_lb_pool_member
AWS::ElasticLoadBalancingV2::Listener	Yes	No	cmdb_ci_lb_service

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
AWS::EC2::NatGateway	Yes	No	cmdb_ci_nat_gateway
AWS::EC2::VPC	Yes	Yes	cmdb_ci_network
AWS::EC2::NetworkAcl	Yes	No	cmdb_ci_network_acl
AWS::EC2::NetworkInterface	Yes	Yes	cmdb_ci_nic
AWS::EC2::RouteTable	Yes	No	cmdb_ci_route_table
AWS::EC2::VolumeAttachment	Yes	Yes	cmdb_ci_storage_mapping
AWS::EC2::Volume	Yes	Yes	cmdb_ci_storage_volume
AWS::EC2::VPNGateway	Yes	No	cmdb_ci_virtual_pvt_gateway
AWS::EC2::VPNConnection	Yes	No	cmdb_ci_vpn_connection
AWS::WAFv2::WebACL	Yes	N/A	cmdb_ci_web_acl
AWS::WAFv2::WebACL	N/A	Yes	cmdb_ci_resource
Tags	Yes	Yes	cmdb_key_value

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
Kubernetes DaemonSet	No	Yes	cmdb_ci_kubernetes_daemonset
Kubernetes deployment	No	Yes	cmdb_ci_kubernetes_deployment
Kubernetes namespace	No	Yes	cmdb_ci_kubernetes_namespace
Amazon EKS node	No	Yes	cmdb_ci_kubernetes_node
Amazon EKS pod	No	Yes	cmdb_ci_kubernetes_pod
Amazon EKS service	No	Yes	cmdb_ci_kubernetes_service
Kubernetes Volume	No	Yes	cmdb_ci_kubernetes_volume
AWS::SSM::ManagedInstanceInventory	No	Yes	cmdb_ci_spkg
AWS::EC2::VPCPeeringConnection	Yes	No	cmdb_aws_elastic_compute_cloud_vpc_peering_connection
AWS::ElasticComputeCloud::VPCEndpointService	Yes	No	cmdb_aws_elastic_compute_cloud_vpc_endpoint_service

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
AWS::Elastic Compute Cloud::EBS Snapshot	Yes	No	cmdb_aws_elastic_cloud_compute_ebs_snapshot
AWS::Elastic Compute Cloud::Reserved Instance	Yes	No	cmdb_aws_elastic_cloud_compute_reserved_instance
AWS::RDS::DBSnapshot	Yes	No	cmdb_aws_relational_database_services_db_snapshot
AWS::SageMaker::Training Job	Yes	No	cmdb_aws_sagemaker_training_job
AWS::Elastic File System::File System	Yes	No	cmdb_aws_elastic_file_system_file_system
AWS::Redshift Serverless::Namespace	Yes	No	cmdb_aws_redshift_serverless_namespace
AWS::Redshift Serverless::Workgroup	Yes	No	cmdb_aws_redshift_serverless_workgroup
AWS::Redshift Serverless::Snapshot	Yes	No	cmdb_aws_redshift_serverless_snapshot
AWS::ElastiCache::Snapshot	Yes	No	cmdb_aws_elasticache_snapshot

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
AWS::CloudTrail::Trail	Yes	No	cmdb_aws_clou dtrail_trail
AWS::Batch::Co mputeEnvironm ent	Yes	No	cmdb_aws_bat ch_compute_en vironment
AWS::StorageGa teway::FileShare	Yes	No	cmdb_aws_stor age_gateway_fil e_share
AWS::DataSync:: Task	Yes	No	cmdb_aws_dat async_task
AWS::API Gateway::Dom ain Name	Yes	No	cmdb_aws_api_ gateway_domai n_name
AWS DynamoDB Cluster	Yes	No	cmdb_ci_cluster _node
AWS::GlobalAcc elerator::Accele rator	Yes	No	cmdb_aws_glob al_accelerator_ accelerator
AWS::Timestrea m::InfluxDBInstan ce	Yes	No	cmdb_aws_time stream_for_influx db_database_in stance
AWS::CloudHSM: :HSM	Yes	No	cmdb_aws_clou dhsm_hsm
AWS::MWAA::En vironment	Yes	No	cmdb_aws_man aged_workflows _for_apache_air

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
			flow_environment
AWS::NetworkFirewall::Firewall	Yes	No	cmdb_aws_network_firewall_firewall

Life cycle management of records in Service Graph Connector for AWS

Life cycle management in the Service Graph Connector for AWS monitors and updates the statuses of AWS resources throughout their entire life cycle, from creation to deletion.

The life cycle management process helps maintain the accuracy and integrity of data in the Configuration Management Database (CMDB).

In life cycle management, the record removal process involves systematically deleting obsolete or unnecessary resources. This step ensures that outdated entries are cleared, keeping the CMDB accurate and up-to-date. See [Record removal process in Service Graph Connector for AWS](#).

Life cycle management for CIs in Service Graph Connector for AWS

The following table lists the configuration items (CIs) in CMDB and other non-CMDB tables for which life cycle management is available in Service Graph Connector for AWS.

Supported tables for life cycle management

Data source	CMDB CI classes	Life cycle management available
SG-AWS-Organization	Cloud Organizations [cmdb_ci_cloud_org]	Yes

Data source	CMDB CI classes	Life cycle management available
SG-AWS-Org-Units	AWS Organizational Unit [cmdb_ci_aws_org_unit]	Yes
SG-AWS-Service-Account	Cloud Service Account [cmdb_ci_cloud_service_account]	Yes
SG-AWS-Service-Account-Tags	Key Value [cmdb_key_value]	No
SG-AWS-Org-Unit-Accounts	None	No
SG-AWS-Datacenters	AWS Datacenter [cmdb_ci_aws_datacenter]	No
SG-AWS-VPC	Network Adapter [cmdb_ci_network_adapter]	Yes
SG-AWS-Subnets	Availability Zone [cmdb_ci_availability_zone]	No
	Cloud Subnet [cmdb_ci_cloud_subnet]	Yes
SG-AWS-Network-Interface	Cloud Mgmt Network Interface [cmdb_ci_nic]	Yes
SG-AWS-Security-Group	Compute Security Group	Yes

Data source	CMDB CI classes	Life cycle management available
	[cmdb_ci_compute_security_group]	
SG-AWS-Storage-Volume	Storage Volume [cmdb_ci_storage_volume]	Yes
	Storage Volume Snapshot [cmdb_ci_storage_volume_snapshot]	Yes
SG-AWS-Image-Private	Image [cmdb_ci_os_template]	Yes
SG-AWS-EC2	Virtual Machine Instance [cmdb_ci_vm_instance]	Yes
	Server [cmdb_ci_server]	Yes
	Block Endpoint [cmdb_ci_endpoint_block]	Yes
	VNIC Endpoint [cmdb_ci_endpoint_vnic]	Yes
	Storage Mapping [cmdb_ci_storage_mapping]	Yes
	Image [cmdb_ci_os_template]	Yes

Data source	CMDB CI classes	Life cycle management available
	IP Address [cmdb_ci_ip_address]	Yes
SG-AWS-Image-Id	Image [cmdb_ci_os_template]	Yes
SG-AWS-Hardware-Type	Hardware Type [cmdb_ci_compute_template]	Yes
SG-AWS-VM-Hw-Consolidation	None	No
SG-AWS-ELB-V1	Cloud Load Balancer [cmdb_ci_cloud_load_balancer]	Yes
SG-AWS-ELB-V2	Cloud Load Balancer [cmdb_ci_cloud_load_balancer]	Yes
SG-AWS-RDS	Cloud DataBase [cmdb_ci_cloud_database]	Yes
SG-AWS-S3	Cloud Object Storage [cmdb_ci_cloud_object_storage]	Yes
SG-AWS-DynamoDb	DynamoDB Table [cmdb_ci_dynamodb_table]	Yes
SG-AWS-API-Gateway	Cloud Gateway [cmdb_ci_cloud_gateway]	Yes

Data source	CMDB CI classes	Life cycle management available
SG-AWS-Lambda	Cloud Function [cmdb_ci_cloud_function]	Yes
SG-AWS-Software-Inventory	Software Installation [cmdb_sam_sw_install]	No
	Software [cmdb_ci_spkg]	No
	Software Instance [cmdb_software_instance]	No
SG-AWS-Software-Remove	None	No
SG-AWS-SSM-SendCommand	TCP Connections [cmdb_tcp]	No
	Running [cmdb_running_process]	No
	Application [cmdb_ci_appl]	No
SG-AWS-Tags	Key Value [cmdb_key_value]	No
SG-AWS-EKS-Cluster	Kubernetes Cluster [cmdb_ci_kubernetes_cluster]	Yes
SG-AWS-EKS-Cluster-2	None	No
SG-AWS-EKS-FULL	Kubernetes Namespace	Yes

Data source	CMDB CI classes	Life cycle management available
	[cmdb_ci_kubernetes_namespace]	
	Kubernetes Node [cmdb_ci_kubernetes_node]	Yes
	Kubernetes Service [cmdb_ci_kubernetes_service]	Yes
	Kubernetes Pod [cmdb_ci_kubernetes_pod]	Yes
	Docker Container [cmdb_ci_docker_container]	Yes
	Kubernetes Volume [cmdb_ci_kubernetes_volume]	Yes
	Kubernetes Deployment [cmdb_ci_kubernetes_deployment]	Yes
	Kubernetes DaemonSet [cmdb_ci_kubernetes_daemonset]	Yes
	Kubernetes ReplicaSet [cmdb_ci_kubernetes_replicaset]	Yes
SG-AWS-Generic-Resources	Cloud Resource [cmdb_ci_cmp_resource]	Yes

Data source	CMDB CI classes	Life cycle management available
SG-AWS-Redshift-Cluster	Amazon Redshift [cmdb_ci_aws_redshift]	Yes
SG-AWS-Generic-Tags	Key Value [cmdb_key_value]	No
SG-AWS-Get-Inventory	Server [cmdb_ci_server]	Yes

Record removal process in Service Graph Connector for AWS

The Service Graph Connector for AWS uses the optimized deletion approach to manage removal of records.

Record removal is the process of handling data that is no longer needed. For any discovered resources deleted later after pulling data, the Service Graph Connector for AWS automatically updates the **Install Status** field of the associated CMDB CI classes to indicate any retired or deleted records.

Deletion of records in Service Graph Connector for AWS involves marking the install status of a record as **retired** rather than permanently deleting it from the system. The record remains in the database, making it possible to reference or restore it later. Deletion for Service Graph Connector for AWS relies on the Source [sys_object_source table], which can become inefficient with large datasets.

To enhance the deletion process, you can enable the optimized deletion method that leverages the SG-AWS Extension Attributes [sn_aws_extension_attributes] table. This table is regularly purged of obsolete records, improving performance and efficiency. The process still marks the install status of a record as **retired** but the streamlined data structure reduces overhead, ensuring quicker processing and better system performance. See [Enable optimized deletion of retired CIs in AWS](#).

Optimized deletion is enabled for all tables in the Service Graph Connector for AWS, except for the following tables:

- Hardware Type [cmdb_ci_compute_template]
- Availability Zone [cmdb_ci_availability_zone]
- AWS Datacenter [cmdb_ci_aws_datacenter]
- Running Process [cmdb_running_process]
- TCP Connections [cmdb_tcp]
- Application [cmdb_ci_appl]